COURSE BOOK B.TECH. III YEAR

(Autonomous)

Computer Science Engineering (CSE)

KIET GROUP OF INSTITUTIONS

www.kiet.edu

Delhi-NCR, Ghaziabad

Connecting Life with Learning

CELEBRATING 25 YEARS OF EXCELLENCE



CURRICULUM STRUCTURE & SYLLABUS

Effective from the Session: 2026-27



VISION

To become a leading institution nationally in the area of professional education, research & innovation for serving the global community.

MISSION

- To impart quality professional education, skills and values through outcome-based innovative teaching learning process in all spheres.
- To undertake collaborative interdisciplinary research as a co-requisite for professional education and simultaneously solve problems faced by society and industry.
- To create an ambience of innovation, entrepreneurship and consultancy for future leaders and innovators.
- To keep faculty members enthusiastic by continuous professional development and positive working environment.

Department of Computer Science Engineering

VISION

To excel in professional education, research, and innovation in the domain of Computer Science and Engineering, fostering a sustainable and inclusive ecosystem for the society.

MISSION

- To inculcate professional skills and knowledge ensuring contribution towards industry and research sustainably.
- To impart skill based education by contemporary tools and techniques.
- To create a conducive environment for academic, research and innovation for providing sustainable solutions.
- To continuously motivate students through an innovative ecosystem that supports lifelong learning and adaptation to technological changes.

B.Tech. Computer Science Engineering

PROGRAM EDUCATIONAL OBJECTIVE

Graduates will be able to:

- Apply the knowledge of computing skills professionally or shall take up higher studies or entrepreneurship.
- Able to design and develop real-time applications ensuring efficient and sustainable computing solutions.
- Exhibit professional ethics, social responsibility, and an understanding of sustainable practices in computing and technology-driven solutions.

PROGRAM OUTCOMES

Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and

inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability fori) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Apply competitive coding techniques efficiently for solving problems using high level languages.

PSO2: Enhance problem-solving and project management skills to align with industry standards in cloud computing and AI-driven applications.

B.Tech. Computer Science Engineering

Teaching Scheme of B. Tech. CSE III Year

B.Tech (CSE) 5th Sem

S No.	Course Category (AICTE)	Course Category (UGC)	BOS	Course Code	Course Name	Type	Academic Learning (AL)			Continuous Internal Examination (CIE)			End Sem Examination (ESE)	Total Marks	Total Credits
							L	T	P	MSE	CA	TOTAL		CIE+ ESE	
1	PC	Major (Core)	CS	CS310L	Theory of Computation	L	3	0	0	60	15	75	75	150	3
2	PC	Major (Core)	CSE	CS311L	Software Engineering and Design Principles	L	3	0	0	60	15	75	75	150	3
3	HS	AEC	ASH	HS203L	Aptitude - 3	L	1	0	0	-	25	25	25	50	1
4	HS	AEC	ASH	HS204L	Soft Skills for Employability - 1	L	2	0	0	-	100	100	-	100	2
Blended															
5	PC	Major (Core)	CSIT	IT401B	Advance Databases	B	3	0	2	80	20	100	100	200	4
6	PC	Major (Core)	CS	CS312B	Deep Learning and Gen AI	B	2	0	2	60	15	75	75	150	3
7	PE	Major (Core)	-	-	Professional Elective – II	B	2	0	4	80	20	100	100	200	4
Lab/Practical															
8	PC	Major (Core)	CSE	CS311P	Software Engineering and Design Principles Lab	P	0	0	2	-	25	25	25	50	1
9	PC	SEC	CS	CS313P	Problem Solving Lab-1	P	0	0	4	-	100	100	-	100	2
10	PW	Summer Internship	CSIT	IT304P	Project Internship	P	0	0	0	-	50	50	-	50	1
Total Hours : 30 hrs.							**16**	**0**	**14**					**1200**	**24**

B.Tech (CSE) 6th Sem

S No.	Course Category (AICTE)	Course Category (UGC)	BOS	Course Code	Course Name	Type	Academic Learning (AL)			Continuous Internal Examination (CIE)			End Sem Examination (ESE)	Total Marks	Total Credits
							L	T	P	MSE	CA	TOTAL		CIE+ ESE	
1	PC	Major (Core)	CSE	CS314L	Natural Language Processing	L	3	0	0	60	15	75	75	150	3
2	PC	Major (Core)	CSE	CS315L	Cryptography and Network Security	L	3	0	0	60	15	75	75	150	3
3	PC	Major (Core)	CSE	CS316L	Image Processing and Computer Vision	L	3	0	0	60	15	75	75	150	3
4	HS	AEC	ASH	HS205L	Aptitude - 4	L	1	0	0	-	25	25	25	50	1
5	OE	Multidisciplinary	-	-	Open Elective-I	L	2	0	0	40	10	50	50	100	2
Blended															
9	PE	Major (Core)	-	-	Professional Elective – III	B	2	0	4	80	20	100	100	200	4
Lab/Practical															
10	HS	AEC	ASH	HS206P	Soft Skills for Employability - 2	P	0	0	2	-	50	50	-	50	1
11	PC	SEC	CS	CS317P	Problem Solving Lab-2	P	0	0	2	-	50	50	-	50	1
12	PW	Research Project / Dissertation	CSIT	IT305P	Minor Project Work	P	0	0	6	-	75	75	75	150	3
Total Hours : 28 hrs.							**14**	**0**	**14**					**1050**	**21**

Professional Electives (PE) – Offered to CSE Students

Course Type (PE)	**Full Stack Product Engineering with AI** *(Powered by Xebia)*	**Next-Gen AI: From Models to Agents** *(Powered by Xebia)*	**DevOps and Autonomous Engineering** *(Powered by Xebia)*	**Data Engineering** (Powered by AWS)	**Applied Apple Technologies: Swift, XCode & iOS Development** *(Powered by Apple)*	**Cyber Security** (Powered by NSDC)
BOS	**CSE**	**CSE**	**CSE**	**CSE**	**CSE**	**CSIT**
PE II- (5thSem)	Backend Servers & API Development with Databases (CS319E)	AIOps & Model Deployment (CS308E)	AI Powered CI/CD & Containerization (CS305E)	AWS Cloud Architecture and Data Engineering (CS337E)	iOS App Development & Swift Programming (CS322E)	Azure Security Engineer Associate (IT307E)
PE III- (6thSem)	Microservices Architecture & System Design (CS320E)	Agentic AI: Building Intelligent & Self-Learning Systems (CS309E)	Cloud & Infrastructure Management (CS306E)	Advanced Data Engineering & AWS Cloud Solutions (CS338E)	Advanced iOS Frameworks & App Store Deployment (CS323E)	AWS Security Specialty (IT308E)

Open Electives -Offered to CSE Students

BOS	ASH	ECE	EEE	ME	KSOM
Open Elective-I (6th Sem)	Reliability Theory & Maintenance (MA202O)	Introduction to Robotic Systems (EC237O)	AIoT for Smart Energy (EE214O)	Advanced Engineering Materials (ME229O)	Business Strategy (MS203O)

Theory Courses Detail Syllabus – V Sem

Course Code: CS310L	Course Name: Theory of Computation	L	T	P	C
		3	0	0	3

Pre-requisite: Discrete Structures & Theory of Logic

Course Objectives:

1. To understand the concept of abstract computational machines.
2. To understand the formal languages and grammars.
3. To understand the relation between the formal languages, grammars, and computational machines.

Course Outcome: After completion of the course, the student will be able to

1. Understand basic concepts of automata theory and formal languages.
2. Construct finite automata for regular expressions and regular languages.
3. Illustrate regular and context-free grammar for formal languages.
4. Construct the pushdown automata for context-free languages.
5. Explore Turing machines for formal languages.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	-	-	-	-	-	-	-	-	-	1
CO2	3	2	2	-	-	-	-	-	-	-	1
CO3	3	2	1	-	-	-	-	-	-	-	1
CO4	3	2	2	-	-	-	-	-	-	-	1
CO5	3	3	2	-	-	-	-	-	-	-	1

Unit 1	Finite Automata	09 hours

Basic Concepts and Automata Theory: Introduction to Theory of Computation- Automata, Computability and Complexity, Alphabet, Symbol, String, Formal Languages, Deterministic Finite Automaton (DFA)- Definition, Representation, Acceptability of a String and Language, Non-Deterministic Finite Automaton (NFA), Equivalence of DFA and NFA, NFA with ε-Transition, Equivalence of NFA's with and without ε-Transition, Minimization of Finite Automata, Myhill-Nerode Theorem.

Unit 2	Finite Automata and Language	09 hours

FA with output and Regular Expressions & Languages**:** Finite Automata with output- Moore Machine, Mealy Machine, Equivalence of Moore and Mealy Machine, Regular Expressions, Transition Graph, Kleene's Theorem, Finite Automata and Regular Expression- Arden's theorem, Algebraic Method Using Arden's Theorem, Regular and Non-Regular Languages- Closure properties of Regular Languages, Pigeonhole Principle, Pumping Lemma, Application of Pumping Lemma, Decidability- Decision properties.

Unit 3	Formal Languages	09 hours

Regular and Non-Regular Grammars: Regular Grammars-Right Linear and Left Linear grammars, Conversion of FA into Regular grammar, Context-Free Grammar (CFG)-Definition, Derivations, Languages, Derivation Trees and Ambiguity, Simplification of CFG, Normal Forms- Chomsky Normal Form (CNF), Greibach Normal Form (GNF), Chomsky Hierarchy.

Unit 4	Push Down Automata	09 hours

Push Down Automata and Properties of Context-Free Languages: Nondeterministic Pushdown Automata (NPDA)- Definition, Moves, A Language Accepted by NPDA, Deterministic Pushdown Automata (DPDA) and Deterministic Context free Languages (DCFL), Pushdown Automata for Context Free Languages, Context Free grammars for Pushdown Automata, Two stack Pushdown Automata, Pumping Lemma for CFL, Closure properties of CFL, Decision Problems of CFL, CYK membership algorithm.

Unit 5	Turing Machine	09 hours

Turing Machines and Recursive Function Theory: Basic Turing Machine Model, Representation of Turing Machines, Language Acceptability of Turing Machines, Techniques for Turing Machine Construction, Modifications of Turing Machine, Turing Machine as Computer of Integer Functions, Universal Turing machine, Linear Bounded Automata, Church's Thesis, Recursive and Recursively Enumerable language, Halting Problem, Post's Correspondence Problem.

Total Lecture hours	45 hours

Textbook:

1. Introduction to Automata Theory, Languages and Computation, J.E.Hopcraft, R.Motwani, and Ullman. 3rd edition, Pearson Education Asia.
2. Peter Linz, "An Introduction to Formal Language and Automata", Narosa Publishing house.
3. K.L.P. Mishra and N.Chandrasekaran, "Theory of Computer Science : Automata, Languages and Computation", PHI
4. Introduction to languages and the theory of computation, J Martin, 3rd Edition, Tata McGraw Hill.K.Krithivasan and R.Rama; Introduction to Formal Languages, Automata Theory and Computation; Pearson Education, 2009.

Reference Books :

1. Daniel I.A. Cohen, John Wiley, "Introduction to languages and the Theory of Computation".
2. Elements and Theory of Computation, C Papadimitrou and C. L. Lewis, PHI 4. Mathematical Foundation of Computer Science, Y.N.Singh, New Age International.
3. Introduction to the Theory of Computation, Michel Sipser, 2nd Edition, Cengage Learning.

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
60	15	75	150
75			

Course Code: CS311L	Course Name: Software Engineering and Design Principles	L	T	P	C
		3	0	0	3

Pre-requisite: Basic programming knowledge

Course Objectives:

1. To introduce students to software engineering principles and software development life cycle models.
2. To develop understanding of software requirements engineering and requirement modeling techniques.
3. To familiarize students with software design principles, UML modeling, and architectural design strategies.
4. To provide knowledge of software metrics, quality assurance, and software testing techniques.
5. To enable students to design maintainable, scalable, and quality software systems using modern engineering practices.

Course Outcome: After completion of the course, the student will be able to

1. Understand software engineering concepts, SDLC models, and software processes.
2. Analyze and document software requirements using standard techniques.
3. Apply software design principles, UML modeling, and architectural strategies.
4. Evaluate software quality using metrics, testing, and quality assurance techniques.
5. Design software systems using modern design approaches and testing methodologies.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	1	-	1	-	-	-	2
CO2	2	3	2	2	2	-	-	1	2	-	1
CO3	2	2	3	2	3	-	-	1	2	1	1
CO4	2	3	1	3	3	-	1	-	2	-	1
CO5	3	2	3	2	3	1	1	2	2	2	1

Unit 1	Introduction to Software Engineering	09 hours

Introduction To Software Engineering and Its Evolution. Software Crisis and The Need for Systematic Software Development. Characteristics Of Good Software. Overview Of Software Development Life Cycle (SDLC). Comparison Of Software Engineering with Traditional Software Engineering. Software Process Models: Waterfall, Incremental, Spiral, And Agile Models. Role Of Documentation and Standards in Software Engineering. Introduction To CASE Tools. Professional Ethics and Responsibility of Software Engineers.

Unit 2	Software Requirements Engineering	09 hours

Concept Of Software Requirements and Their Importance. Functional And Non-Functional Requirements. Requirement Elicitation Techniques: Interviews, Questionnaires, Observation, Prototyping, FAST And QFD. Requirement Analysis and Specification. Software Requirement Specification (SRS): Structure and Characteristics. Requirement Validation and Management. Use Cases and User Stories. Introduction To Requirement Modeling Using UML Use Case Diagrams.

Unit 3	Software Design Concepts and UML Modeling	09 hours
Basic Concepts of Software Design, Architectural Design, Design Strategies: Function-Oriented Design, Object-Oriented Design, Top-Down Design, Bottom-Up Design. Low-Level Design Concepts, Modularization, Design Structure Charts, Pseudocode and Flowcharts, Coupling and Cohesion (Types and Measures), UML Modeling: Class Diagram, Sequence Diagram, Activity Diagram, State Diagram.		
Unit 4	**Software Metrics, Quality Assurance & Complexity Measures**	**09 hours**
Software Metrics and Measurement, Size-Oriented Metrics, Halstead's Software Science, Function Point (FP) Based Metrics Cyclomatic Complexity, Control Flow Graphs (CFG), Software Quality Assurance (SQA), Verification and Validation, SQA Plans, Software Quality Frameworks: ISO 9001, CMMI, Six Sigma.		
Unit 5	**Software Testing & Maintenance**	**09 hours**
Software Testing Fundamentals: Objectives of Testing, Testing Principles, Error, Fault, and Failure Concepts. Levels of Testing: Unit Testing, Integration Testing, System Testing, Acceptance Testing, Regression Testing. Testing Techniques: Functional Testing (Black Box Testing), Structural Testing (White Box Testing), Performance Testing, Load Testing, Stress Testing. Testing Strategies: Top-Down Testing, Bottom-Up Testing, Test Drivers and Test Stubs. Software Maintenance: Need for Maintenance, Types of Maintenance: Corrective, Adaptive, Perfective, Preventive.		
Total Lecture Hours		**45 hours**

Textbook:

1. Pressman, R. S., & Maxim, B. R., "Software Engineering: A Practitioner's Approach, 9th Edition, McGraw-Hill Education", (Covers SDLC, requirements engineering, design concepts, and architectural principles.)
2. Sommerville, I., "Software Engineering", 10th Edition, Pearson Education.

Reference Books:

1. Booch, G., Rumbaugh, J., & Jacobson, I. The Unified Modeling Language User Guide, 2nd Edition, Addison-Wesley. (Authoritative reference for UML modeling techniques.)
2. Gamma, E., Helm, R., Johnson, R., & Vlissides, J., "Design Patterns: Elements of Reusable Object-Oriented Software", Addison-Wesley. (Classic reference for design patterns.)
3. Sommerville, I., "Software Engineering", Pearson Education (earlier editions may also be used as reference).

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
60	**15**	**75**	**150**
75			

Course Code: HS203L	Course Name: Aptitude - 3	L	T	P	C
		1	0	0	1

Pre-requisite: NIL

Course Objectives:

1. To gain a strong foundation in basic mathematical concepts, and become proficient in performing calculations, solving equations, and manipulating numerical data.
2. To apply quantitative techniques in engineering, managerial, and real-life problem contexts. Through practice and exposure to different types of questions, students will learn time-saving techniques and develop strategies to effectively manage time during exams or real-world scenarios.
3. To prepare for various competitive exams, such as aptitude tests for job placements, college admissions tests for higher education (GATE/CAT/GRE/GMAT) and government entrance exams. The course will provide the necessary knowledge and practice to perform well in these exams.

Course Outcome: After completion of the course, the student will be able to

1. understand fundamental concepts of numbers to solve quantitative aptitude problems.
2. apply percentage concepts to accurately calculate and analyze profit, loss, discounts, and value changes in real-life and business- Apply concepts of Simple Interest and Compound Interest to analyze and solve quantitative and financial problems effectively in academic, real-life, and competitive exam scenarios.
3. apply the principles of ratio, proportion, and mixtures to analyze and solve real-world quantitative problems involving comparisons, allocations, and blending of quantities.
4. apply the principles of Arithmetic's to analyze motion related problems, efficiency and work distribution to evaluate practical real-life science, engineering and quantitative problems with accuracy.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	1	-	1	-	-	-	-	-	-	2
CO2	3	1	-	1	-	-	-	-	-	-	2
CO3	3	1	-	1	-	-	-	-	-	-	2
CO4	3	1	-	1	-	-	-	-	-	-	2

Unit 1	Number Systems-I	3 hours

Types of numbers: (Natural, Whole, Integers, Rational, Irrational, Real), Divisibility rules, LCM & HCF (Methods and Applications), Unit digit & last digits, concepts, Assumptions and Simplifications.

Unit 2	Percentages & Applications	3 hours

Basic percentage concepts, Value changes & comparison, Profit, loss & discount, marked price, cost price, selling price relationships, Simple and Compound Interest.

Unit 3	Ratio & Proportion, Averages & Mixtures	3 hours

Ratio & Proportion (Definition of Ratio and Proportion. Type of Proportion (Direct Proportion, Inverse Proportion, Continued Proportion), Definition of Variance (Direct and Inverse), Problem related to age, coins, expenditure and saving, Partnerships). Averages & Mixtures (Weighted averages, Alligation method, Concentration mixing problems)

Unit 4	Time, Speed and Distance (TSD)	3 hours

Basic concepts, definition and terminology related to TSD, Direct and Indirect relation of TSD, Unit Conversions, Average Speed, Relative Speed, Problem on Trains, Problem on Boat & Stream, Race and Games.

Unit 5	Time & Work	3 hours

Introduction to the Time and Work, Work and Wages, Problems based on Efficiency, Problems based on Time and work, Problems based on Pipe and Cistern.

Total Lecture hours	15 hours

Textbook

1. Booklet provided by the Aptitude Team, Training Division -CRPC, KIET.

Reference Books:

1. “Quantitative Aptitude for Competitive Examinations" by R.S. Aggarwal, S. Chand Publication.
2. "How to Prepare for Quantitative Aptitude for the CAT" by Arun Sharma, TMH Publication.
3. "Magical Book on Quicker Maths" by M. Tyra, Published by BSC Publishing Co. Pvt. Ltd
4. "Quantitative Aptitude Quantum CAT" by Sarvesh K. Verma, Arihant Publication.

Reference Websites:

1. https://www.indianbix.com
2. https://www.prepinsta.com
3. https://www.talentsprint.com
4. https://www.careeride.com
5. https://www.smartkeeda.com

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	25	25	50
25			

Course Code: HS204L	Course Name: Soft Skills for Employability - 1	L	T	P	C
		2	0	0	2

Pre-requisite: Students should have basic knowledge of English grammar and should have cleared the Soft Skills Essentials 1 & 2 of B. Tech 2nd year.

Course Objectives:
To enhance the verbal aptitude of the students by strengthening their proficiency in English grammar, vocabulary, and comprehension skills, enabling them to communicate effectively in academic, professional, and competitive examination contexts.

Course Outcome: After completion of the course, the student will be able to:

1. Apply rules of English grammar and sentence structure to construct clear, accurate, and grammatically coherent sentences.
2. Demonstrate enhanced vocabulary usage.
3. Analyze and interpret written passages to answer questions based on reading comprehension, sentence completion, and critical reasoning.
4. Maximize the ability to compose passages solve verbal ability questions commonly asked in placement and recruitment assessments, thereby enhancing overall employability readiness.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	1	3	-	2
CO2	-	-	-	-	-	-	-	1	3	-	2
CO3	-	-	-	-	-	-	-	1	3	-	2
CO4	-	-	-	-	-	-	-	1	3	-	2

Unit 1	About Grammar Skills	8 hours

Tenses and Concord – Concept Revision followed by Practice Exercises on Moodle, Active and Passive voice, Direct & Indirect Narration – Concept Revision followed by Practice Exercises on Moodle, Determiners & Modifiers – Concept Revision followed by Practice Exercises on Moodle, Articles, Conjunctions, Interjections, and Prepositions – Concept Revision followed by Practice Exercises on Moodle

Unit 2	Vocab Building	7 hours

Parallelism, Sentence Improvement & Completion, Conditional Sentences (Practice Exercises on Moodle) Vocabulary – Concept of roots & Affixes, synonyms & antonyms, Idioms & Phrases, Analogies, One-word substitutions, odd-man out, spell check, phrasal verbs (practice exercises on Moodle) Vocabulary.com - Learn Words - English Dictionary

Unit 3	Reading Ability	7 hours

Reading Comprehension & Cloze tests (Practice Exercises on Moodle) https://www.indiabix.com/ Sentence Rearrangement (Jumbled Sentences/Para Jumbles), Missing Sentences (Practice Exercises on Moodle)

Unit 4	Writing Ability	8 hours

Essentials of Writing: Do's and Don'ts of Writing, Writing a Paragraph and Essay Writing (On Word Document checked by Grammarly and display of some sample paragraphs –Poor and good both), Write & Improve with Cambridge

Total Lecture hours	30 hours

Textbook

1. Lewis, Norman. Word Power Made Easy. Goyal Publishers, New Delhi.
2. Wren, P. C., & Martin, H. High School English Grammar and Composition. S. Chand Publishing, New Delhi.
3. Sharma, Arun, & Upadhyay, Meenakshi. How to Prepare for Verbal Ability and Reading Comprehension. McGraw Hill Education, New Delhi.
4. Aggarwal, R. S., & Aggarwal, Vikas. Objective General English. S. Chand Publishing, New Delhi.

References:

1. https://www.indiabix.com/online-test/verbal-ability-test/
2. https://prepinsta.com/
3. https://www.vocabulary.com/
4. https://www.bbc.co.uk/learningenglish

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
-	100	-	**100**
100			

Course Code: IT401B	**Course Name: Advance Databases**	**L**	**T**	**P**	**C**
		3	**0**	**2**	**4**

Pre-requisite: Database System

Course Objectives:

The course aims to provide students with advanced knowledge of database programming, modern NoSQL database systems, and graph database technologies used in current industry applications. The course enables students to develop procedural database programs using PL/SQL for enterprise-level transaction processing, design and manage document-oriented databases using MongoDB for flexible and scalable data storage, and model and analyze highly connected data using Neo4j graph databases and Cypher query language. The course focuses on database design, query optimization, data modeling, data processing, and real-world database application development to prepare students for modern data-driven software systems and multi-model data architectures.

Course Outcome: After completion of the course, the student will be able to

1. Apply PL/SQL concepts- cursors, stored procedures, packages, and triggers for database applications.
2. Understand the document-oriented database solutions using MongoDB including CRUD operations, schema design, indexing, and query operations.
3. Apply the advanced MongoDB features in modern database applications.
4. Elaborate graph-based data models and interpret relationship queries using Neo4j and Cypher query language for data applications.
5. Use advance database tools for demonstrating mini projects.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	2	2	-	1	2	1	1	2
CO2	3	2	2	1	3	-	1	2	1	-	2
CO3	3	2	2	2	3	-	1	2	2	1	2
CO4	3	3	2	2	3	-	1	2	2	1	2
CO5	3	3	3	3	3	1	2	3	2	2	2

Unit 1	**PL/SQL**	**15 hours**

Introduction to PL/SQL, PL/SQL architecture, advantages of PL/SQL, structure of PL/SQL blocks, variables and constants, PL/SQL data types, anchored data types (%TYPE, %ROWTYPE), operators and expressions, control structures (IF, CASE), iterative statements (LOOP, WHILE LOOP, FOR LOOP), SQL statements in PL/SQL, SELECT INTO statement, transaction control statements, implicit and explicit cursors, cursor attributes, parameterized cursors, cursor FOR loop, records, exception handling, predefined and user-defined exceptions, RAISE and RAISE_APPLICATION_ERROR, stored procedures, functions, parameter modes, packages, package specification and package body, database triggers, types of triggers, trigger events, and trigger restrictions.

Mini Projects:

1. Banking Transaction Management System: To implement database-driven banking operations using PL/SQL procedures, functions, and triggers.

2. Student Result Processing and Grade Automation System: To automate student result calculation and grading using PL/SQL.

Unit 2	MongoDB Fundamentals and Data Modeling	20 hours

Introduction to NoSQL databases, overview of MongoDB architecture, installation and MongoDB environment, BSON document structure, collections and documents, CRUD operations (insert, update, delete, find), query operators, projection, sorting and limiting results, indexing concepts and index types, schema design principles, embedding vs referencing documents, data modeling patterns, working with arrays and nested documents, cursor handling, and basic database administration operations.

Mini Projects:

1. E-Commerce Product Catalog Management: To design and implement a document-based database using MongoDB focusing on CRUD operations, schema design, and indexing.
2. Library Management Document Database: To design a document-oriented database focusing on schema design, embedded documents, CRUD operations, and indexing.

Unit 3	Advanced MongoDB Operations and Performance	20 hours

Aggregation framework and aggregation pipeline stages, advanced querying techniques, text search basics, replication concepts and replica sets, sharding fundamentals, transactions in MongoDB, performance optimization and query profiling, backup and restore concepts, security basics including authentication and authorization, monitoring tools overview, deployment considerations, and introduction to MongoDB integration with applications and modern data-driven systems.

Mini Projects:

1. IoT Sensor Data Analytics System: To implement aggregation pipelines, performance optimization concepts, and large data handling using MongoDB.
2. Social Media Analytics and User Activity System: To implement advanced MongoDB features such as aggregation, replication concepts, transactions, and performance optimization.

Unit 4	Neo4j Graph Database	20 hours

Introduction to graph databases and NoSQL graph model, comparison of relational databases and graph databases, property graph model concepts, nodes, relationships, properties, labels, graph database architecture and Neo4j ecosystem overview, installation and Neo4j tools overview, Cypher query language fundamentals, creating and managing nodes and relationships, MATCH, CREATE, MERGE, DELETE and RETURN clauses, filtering, sorting and aggregation in Cypher, path traversal and pattern matching, graph data modeling techniques, schema design in graph databases, indexing and constraints in Neo4j, graph query optimization basics, data import and export concepts, graph analytics basics, shortest path concept, recommendation system modeling, security and access control basics, backup and recovery concepts, performance and scalability concepts, and real-world applications of graph databases.

Mini Projects:

1. Social Network Relationship Graph: To design and query relationship-based data using Neo4j graph modeling and Cypher queries.
2. Movie Recommendation Graph System: To implement graph analytics and recommendation logic using Neo4j relationships and path traversal.

Total Lecture Hours	75 hours

Textbook:

1. PL/SQL by Example — Benjamin Rosenzweig, Elena Silvestrova (Oracle-based PL/SQL programming reference)
2. MongoDB: The Definitive Guide — Kristina Chodorow, Michael Dirolf
3. Graph Databases — Ian Robinson, Jim Webber, Emil

Reference Books:

1. Learning Neo4j — Rik Van Bruggen
2. MongoDB in Action — Kyle Banker

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: CS312B	Course Name: Deep Learning & Gen AI	L	T	P	C
		2	0	2	3

Pre-requisite: NA

Course Objectives:
To introduce students to the fundamentals of Deep Learning and Generative AI, develop primary understanding of neural networks, representation learning, and generation models, along with essential mathematical foundations.

Course Outcome: After completion of the course, the student will be able to

CO1: Apply the working mechanisms of deep neural networks, including multilayer architectures, backpropagation, and optimization methods to improve model performance.

CO2: Analyze deep learning architectures such as CNNs, RNNs, and Autoencoders for classification and sequence-based tasks using standard performance metrics.

CO3: Evaluate generative AI foundations and execute basic GAN architectures for synthetic data generation.

CO4: Evaluate Transformer architectures, LLM training pipelines, and fundamental prompt engineering techniques.

CO5: Evaluate basic Gen-AI applications, including RAG pipelines and tool integration using the Model Context Protocol (MCP) under ethical safety constraints.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	2	2	-	-	-	-	-	3
CO2	3	3	3	2	3	-	-	1	-	-	3
CO3	2	2	3	2	3	1	-	1	-	-	3
CO4	2	3	3	3	3	1	-	2	-	2	3
CO5	2	3	3	3	3	2	3	2	-	2	3

Unit 1 | Training Neural Networks | 09 hours

Introduction to Deep Learning, scope, and transition from MLP to Deep Neural Networks, Backpropagation algorithm, Optimization techniques: Gradient Descent(GD), Loss Functions: Cross Entropy, MSE, MAE, Regularization Techniques: Dropout, L2, Evaluation Metrics: Accuracy, Precision, Recall, F1, AUC, Hyperparameter tuning

Hands on:

1. Build a Simple MLP : Forward Pass & Loss Calculation, Implement forward propagation manually on small dataset Compute loss using MSE and Cross-entropy
2. Backpropagation Demo: Implement backprop manually for a small network (2-2-1), Observe gradient changes
3. Train NN with SGD and compare With Adam, Compare accuracy and convergence graphs
4. Loss Function Comparison Experiment : Train simple regression model using MSE vs MAE, Observe difference in error plots
5. Hyperparameter Tuning Mini Project, Tune epochs, learning rate, batch size, Document performance impact in table form

Unit 2 | Deep Learning Architectures | 17 hours

Convolutional Neural Networks (CNN) architecture, Kernels, Feature Maps, Pooling layers, Flatten operations, Transfer Learning and Pretrained Models, Recurrent Neural Networks (RNN) basics, LSTM and its architecture, Autoencoders

Hands on:

1. Build a Basic CNN for MNIST : Convolution → Relu → Pool → Flatten → Dense
2. Feature Map Visualization :Visualize filters & feature maps from CNN layers
3. Transfer Learning with Pretrained Model
4. Build a Simple RNN, LSTM for Text Classification : Sentiment analysis on IMDB dataset or small CSV
5. Autoencoder for Feature Compression: Build denoising autoencoder on MNIST images

Unit 3	Basics of Generative AI	17 hours

Introduction to Generative Models, Generative vs Discriminative models, Generative Adversarial Networks (GANs) basic architecture, Working of GANs, Types of GANs, AI Ethics and Bias, MCP

Hands on:

1. Generate Random Noise Samples: Use NumPy to generate random vector distribution
2. Mini GAN Concept Simulation : Build a toy GAN on MNIST (low epochs)
3. Bias and Hallucination Experiment: Test AI text generation with biased prompts, observe results
4. Implementation of MCP for connecting LLM and App.

Unit 4	Large Language Models & Gen-AI Applications	17 hours

Transformer architecture and its working, Transformer-based Models: BERT, LLM training pipeline (data, tokenization, fine-tuning), Prompt Engineering Techniques, Retrieval Augmented Generation (RAG) concept, Gen-AI in real applications: text, image, speech, code

Hands on:

1. Implement Simple Transformer Encoder Block :Self-attention layer forward pass demonstration
2. Create a Basic Prompt-Based Chatbot Using OpenAI API/HuggingFace: Simple Q/A chatbot
3. Mini-RAG Pipeline: Upload PDF → Embed text → Ask questions using Retrieval
4. Fine-Tune a Small Language Model :Fine-tune GPT on small custom dataset
5. Image/Text/Code AI Application

Choose one mini-project:

1.Text summarizer 2. Image caption generator 3. Code auto-completion using LLM

Total Lecture Hours	60 hours

Textbook:

1. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, —Deep Learning (Adaptive Computation and Machine Learning), MIT Press
2. Aurélien Géron, —Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media
3. François Chollet, —Deep Learning with Python, Manning Publications

Reference Books:

1. David Foster, —Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play, O'Reilly Media
2. Lewis Tunstall, Leandro von Werra, and Thomas Wolf, —Natural Language Processing with Transformers, O'Reilly Media
3. Chris Fregly and Antje Barth, —Generative AI on AWS: Design, Build, and Deploy at Scale, O'Reilly Media
4. Eugene Charniak, —Introduction to Deep Learning, MIT Press

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
60	15	**75**	**150**
75			

Professional Elective-II Courses Detail Syllabus – V Sem

Course Code: CS319E	Course Name: Backend Servers & Api Development with Databases	L	T	P	C
		2	0	4	4

Pre-requisite: Frontend Engineering with React & Next

Course Objectives:

1. To understand the fundamentals of backend server development
2. To design and develop RESTful APIs
3. To explore database concepts and NoSQL approaches
4. To implement database operations using MongoDB
5. To develop secure and scalable backend applications

Course Outcome: After completion of the course, the student will be able to

1. Apply Node.js runtime concepts, modules, events, streams, and file system operations to develop server-side applications.
2. Develop RESTful APIs using Express.js by applying routing, HTTP methods, request-response handling, and REST design principles.
3. Design structured backend applications using layered architecture, middleware, validation, error handling, and API management techniques.
4. Implement a data access layer by integrating MongoDB with Node.js using Mongoose for data modeling, CRUD operations, relationships, and aggregation.
5. Develop secure backend APIs by implementing authentication, authorization, input validation, and API security mechanisms.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	-	3	-	-	-	-	-	2
CO2	3	3	3	-	3	-	-	2	-	-	2
CO3	3	3	3	2	3	-	-	1	-	-	2
CO4	3	3	3	2	3	-	-	2	1	-	2
CO5	3	2	3	2	3	-	-	2	2	-	2

Unit 1 | Introduction to Node.js | 18 hours

What is Node.js, History of Node.js, Why Node.js, Node.js Architecture, Working and Features, Installation and Setup, Installing Node.js, Launching REPL, Environment, Installing Visual Studio, Code Editor, Components of Node.js, Module Exports, Loading module from, Separate Folder, Modules, File System Module, Reading and Writing into, Files, Appending and Opening Files, Events, In-built Modules File Systems, Operating System, Buffers, The Stream Module

Sample Problems:

1. Write a Node.js program to:
   a) Read a file asynchronously
   b) Append data to an existing file
   c) Display system information using OS module
2. Implement:
   a) EventEmitter to log user login events
   b) Custom module export and import

Unit 2 | REST API & Express.js | 18 hours

Explain REST API, Describe Node.js express, Discuss the importance of express, Explain the installing process of express, Learn express request and response, Describe routing, REST API: Intro to API, History of API Development, Development of AJAX, CRUD

Sample Problems:

1. Develop an Express application to:
   a) Create REST APIs for user data
   b) Perform CRUD operations

2. Implement:
   a) Middleware for logging requests
   b) Error handling middleware

Unit 3	Introduction to Database & NoSQL Approach	18 hours

What is Database, Introduce Database Engineering, Database categories, Concept of NoSQL? Why NoSQL, Compare NoSQL, and RDBMS, NoSQL and SQL, Types of NoSQL Database, Database Properties (NoSQL), Limitation of NoSQL, what is the NoSQL approach? Why Use the NoSQL Approach? Benefits of NoSQL, Types of Databases, Key-Value Stores, Wide-column Stores/ Columnar Databases, Document/Documentstore/Document oriented Databases, Graph-based Database (Already Available with Big Data Theory), Sharding in NoSQL)

Sample Problems:

1. Compare:
   a) SQL vs NoSQL
   b) Advantages of NoSQL
2. Design:
   a) A key-value database model
   b) A document-based data structure

Unit4	Introduction to MongoDB	18 hours

Introduction to MongoDB, MongoDB Module, Features of MongoDB Module, Overview of MongoDB, Principles & Design Goals for MongoDB Server and Database, MongoDB tools, MongoDB Installation on Windows, and Cloud, CRUD operations, Basic MongoDB Commands

Sample problems:

1. Create a simple in-memory data store and then migrate it to MongoDB to persist data.
2. Connect Node.js to MongoDB, perform CRUD operations, and verify results using MongoDB Compass.
3. Query nested documents and arrays in MongoDB using comparison and logical operators ($eq, $in, $and, $or).
4. Build APIs that filter, sort, and paginate data from MongoDB collections.

Unit5	Read & Query Operations	18 hours

Nested documents, Arrays in MongoDB, Sorting Documents, Mongo Shell / Driver, Query Comparison Operators, Nested Documents, Matching an embedded document, Query on Nested Field, Setting up filters using, query operators, Arrays in MongoDB, Querying on Array, Querying the array for an Element, Querying for an Element, using operator, Querying to meet multiple criteria. Nested Arrays in MongoDB, Querying on Nested Arrays, Querying on Array of Embedded Documents, Querying with multiple, conditions on nested Fields, Projections Operations.

Sample problems:

1. Write queries to:
   a) Filter documents using multiple conditions
   b) Query nested arrays

2.Build APIs that filter, sort, and paginate data from MongoDB collections.

3.Secure your API by implementing JWT-based authentication and route protection middleware in Express.

Total Lecture hours	90 hours

Textbook

1. Beginning Node.js, Express & MongoDB Development — Greg Lim
2. Web Development with Node and Express — Ethan Brown
3. Get Programming with Node.js — Jonathan Wexler

Reference Books

1. Node.js Design Patterns — Mario Casciaro
2. Practical Node.js — Azat Mardan

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: CS308E	Course Name: AIOps & Model Deployment	L	T	P	C
		2	0	4	4

Pre-requisite: Machine Learning Fundamentals, Python Programming, Linux Basics, Cloud Computing Concepts

Course Objectives:

1. To introduce students to AIOps concepts and intelligent automation frameworks for modern IT operations.
2. To develop skills in designing and managing automated data pipelines and scalable ML model infrastructure.
3. To enable students to deploy machine learning and LLM models using containerization and orchestration tools.
4. To equip students with knowledge of observability, monitoring, and continuous optimization of AI systems.
5. To explore advanced enterprise-level LLMOps, security, governance, and cost-efficient AI deployment strategies.

Course Outcome: After completion of the course, the student will be able to

1. Understand AIOps concepts and apply AI/ML techniques for anomaly detection, root cause analysis, and predictive maintenance in IT operations.
2. Design and implement automated data pipelines using Kafka, Airflow, Spark, and feature stores for scalable AI/ML workflows.
3. Deploy machine learning and LLM models using serving frameworks, Docker, Kubernetes, and various cloud deployment strategies.
4. Set up observability and monitoring pipelines using Prometheus, Grafana, and ELK Stack to detect model drift and automate retraining.
5. Implement LLMOps frameworks and apply security, compliance, and cost optimization practices for enterprise-grade AI deployment.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	-	2	-	-	-	-	-	2
CO2	3	3	3	2	3	-	-	2	-	-	2
CO3	3	3	3	2	3	-	1	2	-	-	2
CO4	2	3	2	2	2	-	-	2	1	-	2
CO5	2	2	2	-	2	1	1	2	1	-	2

Unit 1 — Introduction to AIOps and Intelligent Automation — 18 hours

Understand AIOps: concept, architecture, and evolution from DevOps to AIOps. Role of AI in IT operations: anomaly detection, root cause analysis, and predictive maintenance. AIOps lifecycle: data ingestion to analytics to correlation to automation. Overview of AIOps platforms: Moogsoft, Dynatrace, Splunk, IBM Watson AIOps. Build a simple log anomaly detector using machine learning.

Sample Problems:

1. Collect and analyze system logs using Python or shell scripts to detect anomalies, and visualize insights using Matplotlib or Grafana.
2. Automate IT operations by building a rule-based or ML-driven script that detects and restarts failed services automatically.

Unit 2 — Data Pipelines and Model Infrastructure — 18 hours

Design of automated data pipelines for AI/ML workflows. Real-time vs batch data processing with Kafka, Airflow, and Spark. Feature store concepts (Feast, Hopsworks) and scalable data management. Model training at scale: TensorFlow Extended (TFX) and Kubeflow Pipelines. Versioning datasets and models using DVC or MLflow Tracking.

Sample Problems:

1. Create an automated data ingestion pipeline using Apache Kafka or Airflow to continuously collect and preprocess operational data.
2. Implement a feature store using Feast or Hopsworks and version datasets using DVC or MLflow Tracking

Unit 3 — Model Deployment Strategies — 18 hours

End-to-end model deployment architecture for ML and LLM systems. Serving frameworks: TensorFlow Serving, TorchServe, BentoML, FastAPI. Containerization and orchestration: Docker, Kubernetes, Helm. Model registry, version control, and rollout strategies: blue-green, canary, and shadow deployments. Deploying LLMs via APIs: OpenAI API, Hugging Face Inference Endpoints, vLLM, Ollama.

Sample Problems:

1. Automate an end-to-end ML workflow using Apache Airflow or Kubeflow for data ingestion, model training, and deployment.
2. Deploy a trained ML model as an API endpoint using Flask or FastAPI and test real-time predictions via REST calls.

Unit4	Observability, Monitoring, and Continuous Optimization	18 hours

Monitoring model performance: accuracy, drift, and fairness. Setting up observability pipelines with Prometheus, Grafana, and ELK Stack. AIOps-driven alerting and root cause correlation. Auto-retraining triggers and feedback loops. Integration with CI/CD workflows: GitHub Actions, Jenkins, GitLab CI/CD.

Sample Problems:

1. Integrate Prometheus and Grafana to monitor deployed model performance, latency, and drift metrics.
2. Build a drift detection mechanism that triggers automatic model retraining using Airflow or Kubeflow.

Unit5	Advanced AIOps & LLM Deployment in Enterprise	18 hours

LLMOps and multi-model orchestration frameworks: LangChain, LlamaIndex, CrewAI. Resource optimization: model quantization, pruning, distillation, caching. Security, compliance, and governance in AI deployment: role-based access, model lineage. Cost-efficient scaling using serverless AI: AWS Lambda, Vertex AI, Azure Functions.

Sample Problems:

1. Build an LLMOps pipeline using LangChain or LlamaIndex for vector storage, retrieval-augmented generation (RAG), and response generation. Deploy the pipeline as a serverless function on AWS Lambda or Vertex AI.
2. Develop a capstone AIOps project combining monitoring, anomaly detection, auto-alerting, and automated model rollback or update. Implement role-based access control and model lineage tracking for compliance and governance.

Total Lecture hours	90 hours

Textbook

1. Introducing MLOps: How to Scale Machine Learning in the Enterprise, Mark Treveil et al., O'Reilly Media
2. Designing Machine Learning Systems, Chip Huyen, O'Reilly Media
3. Machine Learning Engineering, Andriy Burkov, True Positive Inc.

Reference Books

1. Kubeflow for Machine Learning: From Lab to Production, Holden Karau et al., O'Reilly Media
2. The Big Book of MLOps, Databricks Inc.
3. Building Machine Learning Powered Applications, Emmanuel Ameisen, O'Reilly Media

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: CS305E	Course Name: AI Powered CI/CD & Containerization	L	T	P	C
		2	0	4	4

Pre-requisite: DevOps Foundations & Version Control (Linux , Github etc)

Course Objectives:

1. To understand and implement modern CI/CD pipelines, containerization, and deployment strategies using industry tools like Jenkins, Docker, and Kubernetes.
2. To integrate AI-driven automation, security (DevSecOps), and infrastructure management for building scalable and reliable software delivery systems

Course Outcome: After completion of the course, the student will be able to

1. To design and implement end-to-end CI/CD pipelines using tools like GitHub Actions and GitLab CI.
2. To gain the ability to automate infrastructure provisioning using Terraform and manage configurations securely.
3. To containerize and deploy applications using Docker and manage multi-container environments
4. To orchestrate and manage scalable systems using Kubernetes with self-healing and auto-scaling capabilities.
5. To implement secure DevSecOps pipelines with monitoring and AI-based insights using tools like Snyk and Prometheus.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	1	3	-	-	1	1	1	2
CO2	2	3	2	2	3	-	-	1	1	2	2
CO3	2	2	3	2	3	1	-	1	1	2	2
CO4	2	2	3	3	3	1	-	2	1	2	2
CO5	2	2	2	2	3	2	2	2	2	3	3

Unit1	Continuous Integration & Build Automation	18 hours

Principles of CI – automated builds, unit testing, feedback loops, Jenkins, GitHub Actions, GitLab CI pipelines, Maven & Gradle for build management, AI-driven test selection & failure prediction (Harness.io AI or custom ML model), Create a Jenkins pipeline with AI-based test prioritization

Sample Problems:

1. Build a CI pipeline in Jenkins or GitHub Actions to automate build, test, and artifact generation for a web app.
2. Configure Maven or Gradle to manage build dependencies and generate versioned artifacts for each release.
3. Use Harness.io or Jenkins ML plugins to predict build failures and provide AI-based recommendations.

Unit2	Continuous Delivery & Deployment Automation	18 hours

CD concepts – blue/green, canary, and rolling deployments, Infrastructure as Code (IaC) with Terraform / Pulumi, Integration of CI/CD with IaC for environment provisioning, Secrets and configuration management (Ansible / Vault), Deploy a multi stage CD pipeline with Terraform automation

Sample Problems:

1. Implement a blue-green deployment strategy to minimize downtime during application upgrades.
2. Create Infrastructure-as-Code templates using Terraform to provision testing and staging environments automatically.

Unit3	Containerization Fundamentals	18 hours

Docker basics: images, containers, volumes, networking, Docker Compose for multi-container apps, Container security scanning (Trivy / Aqua AI), Image signing and versioning best practices, Containerize a microservice and publish to Docker Hub

Sample Problems:

1. Containerize a microservice application using Docker and manage multi-container deployments with Docker Compose.
2. Scan container images for vulnerabilities using Trivy AI and auto-generate remediation reports.

Unit4	Intelligent Orchestration with Kubernetes & AI Operators	18 hours

Kubernetes architecture: pods, deployments, services, Helm charts and YAML manifest automation, AI-assisted auto-scaling using KEDA / Kubeflow, Self-healing workloads and predictive resource allocation, Deploy a self-healing K8s cluster with AI-based autoscaling

Sample problems:

1. Deploy containerized applications on Kubernetes and manage configuration using Helm charts.
2. Configure KEDA or Kubeflow for AI-based auto-scaling based on real-time CPU or traffic predictions.
3. Implement a self-healing workload that detects crashed pods and auto-restores them using Kubernetes health checks.
4. Set up a canary deployment strategy in Kubernetes and monitor rollout performance using Prometheus and Grafana.

Unit5	DevSecOps & Edge Deployments	18 hours

Shift-left security integration in CI/CD, AI-based vulnerability detection (Snyk AI, GitHub Advanced Security), Observability hooks in pipelines (OpenTelemetry, Prometheus), Edge container deployment (NVIDIA Jetson, Azure IoT Edge), Build a secure CI/CD pipeline with AI-based security and auto-rollback

Sample problems:

1. Integrate Snyk AI or GitHub Advanced Security into CI/CD pipelines for proactive vulnerability detection.
2. Enable distributed tracing in your pipeline using OpenTelemetry and analyze pipeline latency
3. Develop an end-to-end CI/CD pipeline that builds, tests, secures, deploys, monitors, and auto-heals containerized applications.

Total Lecture hours	90 hours

Textbook

1. Fundamentals of Devops and Software Delivery: A Hands-On Guide to Deploying and Managing Software in Production
2. Jenkins → https://www.jenkins.io/doc/
3. GitHub Actions → https://docs.github.com/actions
4. GitLab CI → https://docs.gitlab.com/ee/ci/
5. Terraform → https://developer.hashicorp.com/terraform/docs
6. Docker → https://docs.docker.com/

Reference Books

1. The Devops Handbook: How to Create World-Class Agility, Reliability, & Security in Technology Organizations
2. DevOps Handbook: Introduction to DevOps and its impact on Business Ecosystem

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: CS337E	Course Name: AWS Cloud Architecture and Data Engineering	L	T	P	C
		2	0	4	4

Pre-requisite: AWS Foundations and Data Engineering

Course Objectives:

The objective of this course is to advance learners from cloud fundamentals to professional-grade cloud architecture and data engineering competencies using Amazon Web Services (AWS). Students will design resilient, high-availability multi-tier architectures aligned with the AWS Well-Architected Framework, engineer scalable data lake and analytics pipelines, implement enterprise-grade security and governance models, and apply cost optimization strategies — while preparing for the AWS Certified Solutions Architect – Associate (SAA-C03) certification.

Course Outcome: After completion of the course, the student will be able to

1. Design and evaluate cloud architectures using the AWS Well-Architected Framework, applying best practices for reliability, performance, operational excellence, and cost efficiency.
2. Architect scalable, secure, and highly available AWS infrastructure covering compute, networking, storage, and database services for enterprise workloads.
3. Build and optimize production-grade data lake, data warehouse, and analytics pipelines using AWS-native services aligned with modern data engineering standards.
4. Implement resilient data integration, migration, and orchestration workflows that enable seamless movement and transformation of data across heterogeneous systems.
5. Apply advanced security, governance, compliance, and cost optimization strategies to architect and operate cloud-based data platforms meeting regulatory and business requirements.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	-	2	-	-	-	-	-	2
CO2	3	3	2	2	3	1	-	-	-	-	2
CO3	3	3	3	2	3	-	1	-	-	-	2
CO4	3	3	2	3	3	1	-	-	-	-	2
CO5	3	2	2	2	3	2	2	-	-	-	2

Unit 1	Cloud Architecture Principles, Infrastructure Design, and IaC	18 hours

Cloud Architecture Principles and AWS Well-Architected Framework - Six pillars: Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability. Applying architecture best practices and trade-off analysis in real-world design decisions. Well-Architected Tool: conducting reviews, identifying risk areas, and generating improvement plans.

AWS Global Infrastructure and High-Availability Design - Regions, Availability Zones, Local Zones, Wavelength Zones, and AWS Outposts. Designing for fault tolerance: multi-AZ deployments, cross-region replication, and failover strategies. Service quotas, resilience patterns, and infrastructure redundancy across tiers

AWS Identity and Access Management (IAM) – Advanced Architecture - IAM identity types: users, groups, roles, and identity-based vs. resource-based policies. Permission boundaries, service control policies (SCPs), and attribute-based access control (ABAC). AWS Organizations, AWS Control Tower, and multi-account governance architectures

Virtual Private Cloud (VPC) Design and Network Architecture - VPC architecture: subnets, route tables, internet gateways, NAT gateways, and CIDR planning. Security groups vs. Network ACLs: stateful and stateless filtering, layered defence models. VPC peering, AWS Transit Gateway, VPN, and AWS Direct Connect for hybrid architectures

Infrastructure as Code and Deployment Automation - AWS CloudFormation: stacks, nested stacks, stack sets, change sets, and drift detection. AWS CDK: defining cloud infrastructure using programming languages; constructs and stacks. CI/CD integration with CodePipeline, CodeBuild, CodeDeploy for automated infrastructure delivery

Case Study- Design and Deployment of a Highly Available, Multi-Tier Cloud-Native Application using the AWS Well-Architected Framework

Unit 2	Scalable Compute, Serverless, Container, and Network Architecture	18 hours

Scalable Compute Architectures

• EC2 advanced: instance families, placement groups (cluster/spread/partition), Dedicated Hosts, and Spot strategies

• Auto Scaling Groups: target tracking, step and scheduled policies, lifecycle hooks, and warm pools

• Elastic Load Balancing: ALB (content-based routing, sticky sessions), NLB (ultra-low latency), and GWLB (inline appliances)

Serverless and Event-Driven Application Architecture

• AWS Lambda: concurrency, provisioned concurrency, execution environment, and Lambda@Edge

• Amazon API Gateway: REST, HTTP, and WebSocket APIs; throttling, caching, and usage plans

• AWS Step Functions: standard vs. express workflows; orchestrating data engineering microservices

• Amazon EventBridge: event buses, rules, and event-driven integration patterns

Container-Based Architecture on AWS

• Amazon ECS: task definitions, services, Fargate launch type, and service auto scaling

• Amazon EKS: managed node groups, Fargate profiles, and Kubernetes- native patterns on AWS

• Amazon ECR: image scanning, lifecycle policies, and cross-account repository access

Content Delivery and DNS Architecture

• Amazon CloudFront: distributions, origins, cache behaviours, signed URLs, and Lambda@Edge patterns

• Route 53: routing policies (simple, weighted, latency, geolocation, failover, multi-value), health checks

• Global Accelerator vs. CloudFront: architectural trade-offs for data API acceleration

Hybrid and Edge Architecture

• AWS Snow Family (Snowball Edge, Snowcone) for edge data ingestion and offline transfer

• AWS Direct Connect: dedicated and hosted connections, Link Aggregation Groups (LAGs), failover with VPN

• Storage Gateway: file, tape, and volume gateway modes for hybrid data workflows

Case Study - Architecting a Serverless Microservices Platform for a High-Traffic FinTech Application on AWS

Unit 3	Data Lake Architecture, Analytical Storage, and Query Engineering	18 hours

Data Lake Architecture on AWS

• S3 advanced: storage classes, intelligent tiering, lifecycle rules, versioning, replication (CRR/SRR)

• AWS Lake Formation: data lake creation, fine-grained access control, tag-based governance, and cross-account sharing

• Data lake zones: raw (Bronze), cleaned (Silver), curated (Gold) — design patterns and S3 prefix strategies

AWS Glue – ETL and Data Catalog

• Glue Crawlers, Glue Data Catalog, and schema inference from S3, RDS, DynamoDB, and JDBC sources

• Glue ETL jobs: PySpark and Python Shell scripts, DynamicFrames, job bookmarks, and incremental processing

• Glue DataBrew: no-code data profiling, transformation recipes, and data quality rules

Amazon Redshift – Cloud Data Warehouse Engineering

• Redshift architecture: RA3 nodes, managed storage, leader/compute nodes, and slices

• Distribution styles (KEY, ALL, EVEN, AUTO), sort keys, and query optimization strategies

• Redshift Spectrum: querying S3 data lake with external tables; federated queries to RDS/Aurora

Serverless Analytics – Athena and QuickSight

• Amazon Athena: partitioning, columnar formats (Parquet/ORC), Athena Federated Query, and CTAS patterns

• Amazon QuickSight: datasets, SPICE engine, calculated fields, dashboards, and row-level security

• Cost-efficient analytics architecture: Athena + S3 + Glue Catalog as the query fabric **Data Modelling for Analytical Workloads** • Dimensional modelling: star schema, snowflake schema, and data vault patterns for Redshift • Schema evolution, schema-on-read vs. schema-on-write trade-offs in data lake vs. warehouse • Data mesh principles: domain ownership, federated governance, and data products on AWS **Case Study -** Building a Production-Grade Medallion Data Lake and Self-Service Analytics Platform on AWS		
Unit 4	**Resilient Data Pipelines, Database Architecture, and Operational Observability**	**18 hours**
Real-Time Data Streaming Architecture • Amazon Kinesis Data Streams: shards, sequence numbers, retention, and enhanced fan-out consumers • Amazon Kinesis Data Firehose: delivery to S3, Redshift, OpenSearch; dynamic partitioning and data transformation • Amazon MSK (Managed Kafka): cluster configuration, security (mTLS, IAM), MirrorMaker 2 for replication **Relational Databases – High Availability and Read Scalability** • Amazon RDS: Multi-AZ standby vs. Multi-AZ cluster, automated backups, and point-in-time recovery • RDS Read Replicas: cross-region replicas, promotion strategy, and replica lag monitoring • Amazon Aurora: Aurora Serverless v2, Global Database, Aurora parallel query, and Babelfish **NoSQL, Caching, and Purpose-Built Databases** • Amazon DynamoDB: partition key design, GSIs, LSIs, streams, TTL, DAX caching, and global tables • Amazon ElastiCache (Redis/Memcached): caching patterns (lazy loading, write-through), cluster mode • Purpose-built databases: Neptune (graph), Timestream (time series), QLDB (ledger), OpenSearch Service Data Integration, Migration, and Orchestration • AWS Database Migration Service (DMS): full-load and CDC migrations, schema conversion with SCT • AWS DataSync: automated transfer from on-premises NFS/SMB, HDFS, or S3 to AWS storage services • AWS Step Functions: state machines for multi-step data pipeline orchestration, error handling, and retry logic • Amazon MWAA (Managed Airflow): DAG authoring, task dependencies, and pipeline monitoring **Monitoring, Observability, and Operational Resilience** • Amazon CloudWatch: metrics, alarms, composite alarms, Logs Insights, Contributor Insights, dashboards • AWS X-Ray: distributed tracing for Lambda, API Gateway, and data pipeline microservices • AWS Config, CloudTrail, and AWS Systems Manager for operational visibility and automated remediation **Case Study** - Migrating and Modernizing an Enterprise Data Platform from On- Premises to AWS with Zero-Downtime CDC Pipelines		
Unit 5	**Security Architecture, Data Governance, Cost Optimization, and SAA-C03 Exam Preparation**	**18 hours**
Advanced Security Architecture on AWS • AWS KMS: key types (AWS managed, CMK, imported), key policies, grants, and envelope encryption • Amazon Macie: sensitive data discovery (PII, PHI) in S3; automated classification and alert integration • AWS Security Hub, Amazon GuardDuty, and Amazon Inspector: threat detection and vulnerability management **Data Governance and Compliance Frameworks** • Data classification, tagging strategies, and lifecycle governance using Lake Formation and AWS Glue • Regulatory compliance patterns: GDPR, HIPAA, PCI-DSS on AWS — controls, audit logging, and BAAs • AWS Audit Manager and AWS Artifact: continuous evidence collection and compliance reporting **Cost Optimization Strategies for Data Engineering Workloads** • EC2 pricing models: On-Demand, Savings Plans, Reserved Instances, and Spot Instances — right-sizing • S3 cost optimization: storage class analysis, lifecycle transition, and S3 Intelligent-Tiering automation • Redshift, Athena, and Glue cost controls: concurrency scaling, auto- pause, partition pruning, and query cost estimation **SAA-C03 Exam Domain Review** • Domain 1 – Design Secure Architectures (30%): IAM, encryption, VPC security, and DDoS protection • Domain 2 – Design Resilient Architectures (26%): HA, disaster recovery RTO/RPO, decoupling, and caching • Domain 3 – Design High-Performing Architectures (24%): compute, storage, database, and network performance • Domain 4 – Design Cost-Optimized Architectures (20%): pricing models, managed services, and scaling economics **Case** Study - Designing a Secure, Compliant, and Cost-Optimized Cloud Data Platform Architecture for a Healthcare Enterprise on AWS Exam Prep Plan: AWS Certified Solutions Architect – Associate (SAA-C03) Official Practice Exam: AWS Certified Solutions Architect – Associate (SAA-C03)		
Total Lecture hours		**90 hours**

Textbook

1. Ramamoorthy, C. (2023). AWS Certified Solutions Architect Study Guide: Associate SAA-C03 Exam. Sybex / Wiley.
2. Wittig, M., & Wittig, A. (2023). Amazon Web Services in Action (3rd ed.). Manning Publications.

Reference Books

1. Singh, G. (2023). Data Engineering with AWS. Packt Publishing.
2. Tremblay, B. (2022). Fundamentals of Data Engineering. O'Reilly Media.
3. AWS Documentation and Whitepapers: AWS Well-Architected Framework, AWS Data Analytics Lens, and AWS Security Best Practices.

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: CS322E	Course Name: iOS App Development & Swift Programming	L	T	P	C
		2	0	4	4

Pre-requisite: Foundation of iOS App Development

Course Objectives:

1.To develop proficiency in advanced Swift programming and responsive SwiftUI application development.
2. To build data-driven iOS applications using networking, JSON processing, and local data persistence.

Course Outcome: After completion of the course, the student will be able to

CO1: To Apply advanced Swift programming concepts to develop reusable and efficient code.
CO2: To Apply responsive user interfaces using SwiftUI layouts.
CO3: To Develop multi-screen iOS applications with navigation and media integration.
CO4: To Integrate REST APIs and parse JSON data in SwiftUI applications.
CO5: To Develop and debug SwiftUI applications using local data persistence techniques.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	1	2	-	-	-	-	-	2
CO2	2	2	3		3	-	-	-	-	-	2
CO3	2	2	3		3	-	-	-	-	-	2
CO4	3	3	3	2	3	-	-	-	-	-	2
CO5	2	3	3	2	3	-	-	-	-	-	2

Unit 1	Advanced Swift Language Concepts	18 hours

Generics: Writing generic functions and custom types to eliminate redundant code.
Protocol-Oriented Programming: Creating protocols, applying protocol conformance, and using protocol extensions for reusable code.
Memory & Lifecycle Basics: Understanding Automatic Reference Counting (ARC), avoiding simple retain cycles, and managing SwiftUI app states using ScenePhase.

Unit 2	SwiftUI Layouts & Responsive UI	18 hours

Core Layout Containers: Structuring user interfaces cleanly using VStack, HStack, and ZStack.
Adaptive Design: Utilizing GeometryReader to handle view dimensions dynamically.
Screen Adaptability: Best practices for building layouts that scale across different screen sizes.

Unit 3	SwiftUI Navigation & Media Handling	18 hours

Navigation Components: Implementing view transitions using NavigationStack and NavigationLink.
Tab Interfaces: Creating multi-tab architectures using TabView.
Camera & Photo Library: Requesting user permissions, opening the camera/photo library, and displaying selected images in SwiftUI views.

Unit 4	Networking & JSON Parsing	18 hours

Web API Connections: Using URLSession to fetch remote data from simple REST APIs.
Data Conversion: Working with the Codable protocol to map network responses to Swift models.
JSON Handling: Parsing standard JSON data structures and updating the SwiftUI user interface with the results.

Unit 5	Data Persistence & Project	18 hours

UserDefaults: Storing and retrieving small user preferences and application settings.
FileManager: Simple techniques for reading and writing data files to the device storage.
API-Based Mini Project: Developing and debugging a simple, integrated SwiftUI application that fetches network data, displays a dynamic layout, and saves user settings.

Total Lecture Hours	90 hours

Textbook:
1. SwiftUI Essentials – iOS Edition, Neil Smyth, SwiftUI Essentials – iOS Edition, Payload Media, Latest Edition.
2. iOS Programming: The Big Nerd Ranch Guide, Christian Keur and Aaron Hillegass, iOS Programming: The Big Nerd Ranch Guide, Big Nerd Ranch Guides, Latest Edition.

Reference Books:
1. Swift Programming: The Big Nerd Ranch Guide, Matthew Mathias and John Gallagher, Swift Programming: The Big Nerd Ranch Guide, Big Nerd Ranch Guides, Latest Edition.
2. Develop in Swift: Explorations, Apple Education, Develop in Swift: Explorations, Apple Inc., Latest Edition.

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: IT307E	Course Name: Azure Security Engineer Associate	L	T	P	C
		2	0	4	4

Pre-requisite: Cloud Computing Fundamentals, Basic Networking, Basic Cyber Security Concepts

Course Objectives: This course provides in-depth knowledge of Microsoft Azure security technologies aligned with the AZ-500 certification. It focuses on identity and access management, platform protection, security operations, and data security in Azure environments. The course aims to develop practical skills for securing cloud infrastructure, detecting and responding to threats, implementing governance, and preparing students for Azure Security Engineer Associate certification and cloud security roles.

Course Outcome:
CO1: Understand Azure security concepts, Zero Trust principles, and identity protection mechanisms.
CO2: Implement secure identity and access management using Azure Active Directory and RBAC.
CO3: Configure platform protection including network security, compute security, and endpoint protection.
CO4: Apply data security techniques including encryption, key management, and secure storage.
CO5: Monitor, detect, and respond to security threats using Azure security tools and prepare for AZ-500 certification

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	2	-	1	-	-	-	2
CO2	3	3	2	2	3	-	1	-	-	-	2
CO3	3	3	2	2	3	-	1	-	-	-	2
CO4	3	2	2	2	3	-	1	-	-	-	2
CO5	3	3	2	2	3	-	1	-	-	-	2

Unit 1	Identity and Access Management in Azure	18 hours
Introduction to Zero Trust security model, Azure Active Directory (Azure AD), users, groups, roles, and role-based access control (RBAC). Multi-Factor Authentication (MFA), Conditional Access policies, Privileged Identity Management (PIM), identity protection, and secure authentication mechanisms. Case Study: Implementing Secure Access using MFA and Conditional Access Policies in Azure.		
Unit 2	**Platform Protection: Network and Compute Security**	**18 hours**
Azure Virtual Network security, Network Security Groups (NSG), Azure Firewall, Web Application Firewall (WAF), DDoS protection. Securing virtual machines, disk encryption, endpoint protection, and Azure Defender for servers and workloads. Case Study: Designing a Secure Network Architecture using Azure Firewall and NSGs.		
Unit 3	**Security Operations and Threat Management**	**18 hours**
Azure Security Center (Microsoft Defender for Cloud), security posture management, vulnerability assessment. Azure Sentinel (SIEM & SOAR), log collection, threat detection, incident response, and automation using playbooks. Case Study: Threat Detection and Incident Response using Azure Sentinel.		
Unit 4	**Data Security and Key Management**	**18 hours**
Data encryption in Azure (at rest and in transit), Azure Key Vault, secrets, keys, certificates management. Storage security, database security, Transparent Data Encryption (TDE), Always Encrypted, and secure data access. **Case Study: Securing Sensitive Data using Azure Key Vault and Encryption Services.**		
Unit 5	**Governance, Compliance, and Security Management**	**18 hours**
Azure Policy, Blueprints, compliance management, regulatory standards (ISO, GDPR basics). Resource locks, tagging, cost security considerations. Azure Monitor, Log Analytics, and security auditing. **Case Study: Implementing Governance and Compliance using Azure Policy and Blueprints.**		
	Total Lecture Hours	**90 hours**

Textbook:

1. Microsoft – Exam Ref AZ-500 Microsoft Azure Security Engineer Associate
2. Yuri Diogenes – Microsoft Azure Security Technologies (AZ-500)
3. Microsoft Learn – Azure Security Documentation

References:

1. Microsoft Learn – Azure Security Learning Paths
2. https://learn.microsoft.com/en-us/training/
3. AZ-500 Certification Guide

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
80	20	**100**	**200**
100			

Practical Courses Detail Syllabus – V Sem

Course Code: CS311P	Course Name: Software Engineering and Design Principles Lab	L	T	P	C
		0	0	2	1

Pre-requisite: Basic Programming Knowledge

Course Objectives:

- To apply software engineering concepts in practical software development scenarios.
- To train students in software requirement analysis and UML-based design.
- To expose students to modern software development workflows and testing practices.
- To develop professional documentation and collaborative development skills.
- To familiarize students with software quality assurance and software testing tools.

Course Outcome: After completion of the course, the student will be able to

1. Apply software engineering principles to analyze software requirements and prepare professional software documentation.
2. Develop UML models and software designs using standard design principles and architectural strategies.
3. Apply software metrics, testing techniques, and quality assurance practices using modern software tools.
4. Demonstrate collaborative software development using version control, agile practices, and effective project presentation skills.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	3	2	2	2	-	-	1	2	1	1
CO2	2	2	3	2	3	-	-	2	3	1	1
CO3	2	3	2	3	3	-	1	1	2	1	2
CO4	1	1	2	-	3	-	1	3	3	2	2

List Of Practical's (Indicative & Not Limited To)

Session 1: Introduction to Software Engineering Tools

- Introduction To Git And Github
- Repository Creation, Version Control Basics
- Software Project Lifecycle In Industry
- Team Formation And Project Domain Selection

Deliverable: GitHub Repository Setup
Industry Mapping: Collaborative Development

Session 2: Problem Identification & Feasibility Analysis

- Identification of Real-World Software Problems
- Problem Statement Writing
- Technical and Operational Feasibility
- Project Scope Definition

Deliverable: Problem Statement Document
Industry Mapping: Product Requirement Analysis

Session 3: Requirements Engineering – Elicitation

- Stakeholder identification
- Requirement gathering techniques:
  - ➢ Interview simulation
  - ➢ Questionnaire design
  - ➢ User persona creation
- User story writing

Deliverable: Requirement Gathering Report
Industry Mapping: Business Analysis and Client Interaction

Session 4: SRS Documentation

- Preparation Of Software Requirement Specification (SRS)
- IEEE SRS Format:
  - Introduction
  - Overall Description
  - Functional Requirements
  - Non-Functional Requirements

Tools: MS Word / Google Docs / Overleaf
Deliverable: Complete SRS Document
Industry Mapping: Formal Documentation Standards

Session 5: UML Use Case Modeling

- Use Case Identification
- Use Case Diagram Creation

- Use Case Description Templates

Tools: Star UML / Draw.Io
Deliverable: Use Case Diagram And Narratives
Industry Mapping: Requirement Modeling

Session 6: UML- Class Diagram Design

- Object-Oriented Analysis
- Class Identification, Attributes, Methods, Relationships
- UML Class Diagram Modeling

Deliverable: Class Diagram
Industry Mapping: Object-Oriented System Design

Session 7: UML- Sequence and Activity Diagram Modeling

- Sequence Diagram Design
- Message Flow Representation
- Workflow Modeling Using Activity Diagram

Deliverable: Sequence And Activity Diagram
Industry Mapping: Interaction And Workflow Modeling

Session 8: Software Design Principles

- Modularization
- Coupling and Cohesion Analysis
- Function-Oriented vs Object-Oriented Design
- Top-Down and Bottom-Up Design

Deliverable: Design Analysis Report
Industry Mapping: Software Design Engineering

Session 9: Software Architecture Design and Design Patterns

- Layered Architecture Design
- Client-Server
- Microservices (Conceptual)

Deliverable: System Architecture Diagram
Industry Mapping: System Architecture Planning

Session 10: Software Metrics & Complexity Analysis

- Size-Oriented Metrics
- Halstead Metrics Calculation
- Function Point Analysis
- Cyclomatic Complexity
- Control Flow Graphs

Deliverables: Metrics Calculation Report
Industry Mapping: Software Quality Measurement

Session 11: Unit & Integration Testing

- Writing Test Cases
- Unit Testing Concepts
- Integration Testing Strategies
- White Box Testing Basics

Tools: JUnit / Manual Testing Templates
Deliverable: Test Case Document
Industry Mapping: Software Testing Practices

Session 12: Functional & Performance Testing

- Black Box Testing
- Functional Testing
- Load Testing
- Stress Testing
- Regression Testing

Deliverable: Testing Report
Industry Mapping: Quality Assurance Engineering

Session 13: Agile Workflow & Project Management

- Agile Methodology
- Sprint Planning
- Backlog Creation
- Scrum Board Demonstration

Tool: Trello / GitHub Projects
Deliverable: Agile Sprint board
Industry Mapping: Agile Industry Workflow (Scrum model)

Session 14: Final Project Design Review

- Review of SRS
- UML Review
- Architecture Validation
- Testing Documentation Review

Deliverable: Final Design & Testing Report
Industry Mapping: Software Review Process

Session 15: Final Project Demonstration and Viva

- Final Project Presentation
- UML Model Explanation
- Architecture Walkthrough
- Demonstration Of Testing Reports
- SRS + Design Document Submission

Deliverable: Final Project Documentation
Industry Mapping: Client Demo and Technical Presentation

Total Hours	**30 hours**

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
-	**25**	**25**	**50**
25			

Course Code: CS313P	**Course Name: Problem Solving Lab-1**	**L**	**T**	**P**	**C**
		0	**0**	**4**	**2**

Pre-requisite: NIL

Course Objectives:

1. To familiarize students with the basic and advance of Problem Solving.
2. To learn modern techniques to solve DSA Problems.

Course Outcome: After completion of the lab, the student will be able to

1. Apply software engineering principles to analyze software requirements and prepare professional software documentation.
2. Develop UML models and software designs using standard design principles and architectural strategies.
3. Apply software metrics, testing techniques, and quality assurance practices using modern software tools.
4. Demonstrate collaborative software development using version control, agile practices, and effective project presentation skills.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	–	1	–	–	–	–	–	1
CO2	3	3	2	–	2	–	–	–	–	–	1
CO3	2	3	2	1	1	–	–	–	–	–	1
CO4	2	2	3	1	3	–	–	2	1	1	2

List of Experiments:

1. Basic Logic Building - 1. Print all divisors 2. Prime check 3. GCD using Euclid
2. Number Theory Basics - 1. Sieve of Eratosthenes 2. Count primes in range 3. Modular arithmetic problems
3. Recursion-I - 1. Factorial/Power recursion 2. Print subsets 3. Fibonacci recursion
4. Recursion-II/Backtracking Basics - 1. N-Queen (4×4) 2. Rat in a maze 3. Print permutations
5. Two Pointer Technique–I - 1. Pair sum 2. Remove duplicates from sorted array 3. Move zeros to end
6. Two Pointer Technique–II - 1. 3-sum 2. Container with most water 3. String palindrome check
7. Sliding Window–I - 1. Max sum subarray (fixed k) 2. Count anagram occurrences 3. Longest substring without repeat
8. Sliding Window–II - 1. Minimum window substring 2. Longest repeating character replacement 3. Fruit into baskets
9. Binary Search–I - 1. First/last occurrence 2. Square root using BS 3. Search in rotated sorted array
10. Binary Search on Answer - 1. Allocate books 2. Aggressive cows 3. Koko eating bananas
11. Greedy–I - 1. Activity selection 2. Minimum coins 3. Jump game
12. Greedy-II - 1. Job sequencing 2. Gas station problem 3. Minimum platforms
13. Bit Manipulation–I - 1. Count set bits 2. Find unique element using XOR 3. Check power of 2
14. Bit Manipulation–II - 1. Subset generation using bits 2. Swap bits in number 3. Count bits from 1 to N

Lab-1 Mini Project - 1. Pattern-based problem sheet (mix) 2. Contest simulation 3. 10 mixed technique problems

Total Laboratory Hours	**60 hours**

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	100	-	100
100			

Theory Courses Detail Syllabus – VI Sem

Course Code: CS314L	Course Name: Natural Language Processing	L	T	P	C
		3	0	0	3

Pre-requisite: Basic Python, Probability and Machine learning concept

Course Objectives:

1. To introduce fundamental concepts of Natural Language Processing and its linguistic foundations.
2. To enable students to perform text preprocessing and feature extraction techniques.
3. To understand syntactic and semantic analysis of natural language using formal and statistical methods.
4. To develop probabilistic language models and evaluate their performance.
5. To design and implement real-world NLP applications such as information extraction, sentiment analysis and dialogue systems.

Course Outcome: After completion of the course, the student will be able to

1. Understand linguistic concepts and perform text preprocessing using NLP techniques.
2. Apply syntactic analysis and parsing algorithms to analyze sentence structure.
3. Construct and evaluate statistical language models using probabilistic methods.
4. Analyze semantic meaning using word sense disambiguation and representation techniques.
5. Develop NLP applications such as text classification, sentiment analysis and conversational systems.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	1	3	-	-	-	1	-	2
CO2	3	3	2	2	2	-	-	-	-	-	2
CO3	3	3	2	3	2	-	-	-	-	-	3
CO4	3	2	2	2	2	-	-	-	-	-	3
CO5	2	3	3	2	3	1	1	2	2	1	3

Unit 1	Introduction and Text Pre-processing	9 hours

Introduction to Natural Language Processing, applications of NLP, NLP pipeline and components, challenges in NLP. Words and their components, lexemes, morphemes, morphology, morphological typology. Text preprocessing: sentence segmentation, tokenization, normalization, stop word removal, stemming, lemmatization. Part of Speech tagging, chunking, Named Entity Recognition. Feature extraction: Bag of Words, N-grams, Term Frequency–Inverse Document Frequency (TF-IDF).

Unit 2	Syntax and Parsing	9 hours

Representation of syntactic structure, constituency and dependency grammar, treebanks. Parsing natural language: top-down and bottom-up parsing. Shift-reduce parsing, chart parsing, CYK algorithm. Ambiguity in parsing, probabilistic context free grammar (PCFG). Generative and discriminative models for parsing.

Unit 3	Language Modelling	9 hours

Introduction to language models, chain rule and Markov assumption. N-gram models: unigram, bigram and trigram models. Parameter estimation and smoothing techniques: Laplace, Good-Turing, Kneser-Ney. Language model evaluation: entropy and perplexity. Class-based and neural language models (overview).

Unit 4	Semantic Analysis	9 hours

Semantics and meaning representation, lexical semantics. Word sense disambiguation, semantic similarity measures. Semantic parsing and predicate-argument structure. Thematic roles and semantic frames. Vector semantics and word embeddings. Knowledge representation: ontologies and WordNet.

Unit 5	Discourse Processing and Applications	9 hours

Discourse processing, cohesion and coherence. Coreference and anaphora resolution. Dialogue systems and conversational structure. Applications: text classification, sentiment analysis, machine translation and question answering. Introduction to neural NLP: attention mechanism, transformer models. Evaluation metrics: precision, recall and F-measure, ethical issues in NLP.

Total Lecture hours	**45 hours**

Textbook

1. Daniel Jurafsky and James H. Martin, Speech and Natural Language Processing, 3rd Edition (Draft/Latest), Pearson.
2. Daniel M. Bikel and Imed Zitouni, Multilingual Natural Language Processing Applications: From Theory to Practice, Pearson Education.
3. Jacob Eisenstein, Introduction to Natural Language Processing, MIT Press.

Reference Books

1. Tanveer Siddiqui and U.S. Tiwary, Natural Language Processing and Information Retrieval, Oxford University Press.
2. Steven Bird, Ewan Klein and Edward Loper, Natural Language Processing with Python, O'Reilly Media.
3. Yoav Goldberg, Neural Network Methods for Natural Language Processing, Morgan & Claypool.
4. Christopher D. Manning and Hinrich Schütze, Foundations of Statistical Natural Language Processing, MIT Press.
5. Delip Rao and Brian McMahan, Natural Language Processing with PyTorch, O'Reilly Media.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
60	**15**	**75**	**150**
75			

Course Code: CS315L	**Course Name: Cryptography and Network Security**	**L**	**T**	**P**	**C**
		3	**0**	**0**	**3**

Pre-requisite: NIL

Course Objectives:

1. To introduce the fundamental concepts of information security, cryptography, and network security models.
2. To develop an understanding of symmetric and asymmetric cryptographic algorithms and key management techniques.
3. To study data integrity, authentication mechanisms, and digital signature techniques used in secure communication.
4. To understand secure communication protocols and web security mechanisms used in modern networks.
5. To analyze network security architectures such as IPsec, VPNs, and firewall systems for protecting network infrastructure.

Course Outcome: After completion of the course, the student will be able to

1. Explain the fundamental concepts of cryptography, security goals, types of attacks, and classical cryptographic techniques used in information security.
2. Analyze symmetric and public-key cryptographic algorithms, including AES, DES, RSA, and Diffie–Hellman, along with their applications in secure communication.
3. Apply hash functions, message authentication codes, digital signatures, and authentication mechanisms to ensure data integrity and secure communication.
4. Evaluate secure communication protocols and web security technologies such as SSL/TLS, HTTPS, SSH, and firewall systems.
5. Analyze the IPsec architecture, security associations, authentication headers, and VPN mechanisms used to secure IP-based communication networks.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	1	1	1	1	1	-	1	-	1
CO2	3	2	2	2	2	0	1	-	1	-	1
CO3	2	3	2	2	2	1	2	-	1	-	1
CO4	2	2	2	2	3	1	2	-	1	-	1
CO5	2	3	2	2	3	1	1	-	1	1	1

Unit1	**Foundations of Cryptography & Security**	**9 hours**

Introduction to information and network security, Need for security; real-world incidents and impacts, Security goals: confidentiality, integrity, availability, authenticity, accountability, Types of security attacks Security services and mechanisms, Model for network security, Cryptography introduction, Symmetric vs asymmetric cryptography (conceptual overview), Classical substitution and transposition ciphers; limitations, Steganography: concept and simple modern examples

Unit2	Symmetric and Public-Key Cryptography	9 hours

Symmetric key ciphers: design goals, confusion and diffusion, Block cipher principles: Feistel and substitution–permutation structures, DES: structure, key size, weaknesses and deprecation, Triple-DES (brief): role as a legacy enhancement. AES: key sizes, high-level round structure, industry applications, Overview of other symmetric ciphers: Blowfish, RC5, IDEA Block cipher modes, Stream ciphers, RC4 and reasons for its deprecation Public-key cryptography: problem of key distribution, trapdoor one-way functions,RSA algorithm: conceptual key generation, encryption, decryption, signatures, Diffie–Hellman key exchange, Knapsack-style public-key schemes. Suggested Practical Exercise: Applied Cryptography with AES, RSA, and Diffie–Hellman Using Python/OpenSSL

Unit3	Data Integrity & Authentication	9 hours

E-mail security requirements and threats (spoofing, tampering, phishing), Pretty Good Privacy (PGP): keys, web of trust, encrypting and signing Cryptographic hash functions: properties and applications, Secure Hash Algorithm : SHA-2/SHA-512; Why MD5 and SHA-1 are considered broken/insecure. Message Authentication Codes (MACs): purpose and security requirements, HMAC, CMAC Digital signatures: goals (integrity, authentication, non-repudiation), Conceptual view of RSA/ElGamal-style signature schemes Authentication procedures: challenge–response, signed tokens, certificate-based login Case study on password security, key-stretching methods, trusted software updates, and digital code-signing as used in real-world security systems.

Unit4	Secure Protocols & Web Security	9 hours

Email security, S/MIME: certificate-based email security; integration with PKI, Web security considerations: sessions, cookies, basic browser threat landscape Overview of SSL and TLS evolution, TLS handshake: certificate verification, cipher suites, key establishment (conceptual) HTTPS: HTTP over TLS, Secure Shell (SSH): secure remote login, tunneling, key-based authentication, Firewalls: packet-filtering vs stateful, perimeter vs host firewalls, basic rule examples

Unit5	IP Security	9 hours

IP security motivation: limitations of link-layer and application-layer only, IPsec overview and architecture: policies, Security Associations (SAs), Authentication Header (AH): what it protects, use cases, NAT issues (high level), Encapsulating Security Payload (ESP): confidentiality, integrity, anti-replay, Transport mode vs tunnel mode; host-to-host vs site-to-site VPNs, IPsec key management: high-level Internet Key Exchange (IKE) concepts

Total Lecture hours	45 hours

Textbook:

1. William Stallings, “Cryptography and Network Security: Principals and Practice”, Pearson Education.
2. Behrouz A. Frouzan: Cryptography and Network Security, McGraw Hill .
3. C K Shyamala, N Harini, Dr. T.R.Padmnabhan Cryptography and Security ,Wiley

Reference Books:

1. Bruce Schiener, “Applied Cryptography”. John Wiley & Sons
2. Bernard Menezes,” Network Security and Cryptography”, Cengage Learning.
3. Atul Kahate, “Cryptography and Network Security”, McGraw Hill

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
60	15	75	150
75			

Course Code: CS316L	Course Name: Image Processing and Computer Vision	L	T	P	C
		3	0	0	3

Pre-requisite: NIL

Course Objectives:

1. To recall and recognize fundamental concepts of digital image representation, color models, and image formation.
2. To understand and explain spatial and frequency domain filtering techniques for image enhancement.
3. To apply image segmentation and morphological processing algorithms to solve image analysis problems.
4. To analyze and compare feature extraction methods and their suitability for various computer vision tasks.
5. To evaluate deep learning architectures and assess their effectiveness for image classification and object detection.

Course Outcome: After completion of the course, the student will be able to

1. Describe the fundamentals of digital image representation, pixel relationships, color models, and image acquisition processes.
2. Apply spatial and frequency domain filtering techniques to perform image enhancement and noise suppression.
3. Implement image segmentation and morphological algorithms, and analyze their performance on different image types.
4. Analyze and compare image feature descriptors such as Harris, SIFT, ORB, and HOG for computer vision applications.
5. Evaluate the architecture and effectiveness of CNN-based models and object detection frameworks for real-world image understanding tasks.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	-	2	-	1	-	1	-	2
CO2	3	3	2	2	2	-	1	-	1	-	2
CO3	3	3	2	2	2	-	1	1	1	-	2
CO4	3	2	2	1	3	-	1	1	1	-	2
CO5	2	2	2	2	3	1	1	1	2	1	3

Unit 1	Introduction to Digital Image Processing	9 hours

Digital image fundamentals: Define and describe image representation and types (binary, grayscale, color). Explain image acquisition and digitization — sampling and quantization. Describe pixel relationships: neighbors, adjacency, and connectivity. Identify image file formats (BMP, JPEG, PNG) and their properties. Explain color models: RGB, HSV, YCbCr, CMY/CMYK and their differences. Describe intensity transformation functions: log, power-law (gamma), piecewise-linear. Explain histogram processing: interpret histograms and describe histogram equalization and specification.

Unit 2	Spatial and Frequency Domain Filtering	9 hours

Spatial filtering: smoothing filters (mean, median, Gaussian), sharpening filters (Laplacian, Sobel, Prewitt). Unsharp masking and highboost filtering. Frequency domain concepts: Discrete Fourier Transform (DFT) and its properties. Fast Fourier Transform (FFT). Frequency domain filtering: ideal, Butterworth, and Gaussian low-pass and high-pass filters. Band-pass and band-reject filters. Homomorphic filtering.

Unit 3	Image Segmentation, Morphology and Restoration	9 hours

Image segmentation: global and adaptive thresholding, Otsu's method. Edge-based segmentation: Roberts, Prewitt, Sobel, and Canny edge detectors. Region-based segmentation: region growing, region splitting and merging. Morphological image processing: structuring elements, erosion, dilation, opening, closing, hit-or-miss transform. Morphological algorithms: boundary extraction, hole filling, connected components labeling. Introduction to image restoration: noise models (Gaussian, Salt-and-Pepper), mean filters for noise removal, introduction to Wiener filtering.

Unit 4	Feature Extraction and Computer Vision Fundamentals	9 hours

Feature extraction: Harris corner detector, Shi-Tomasi corner detection. Scale-Invariant Feature Transform (SIFT): keypoint detection and description. ORB (Oriented FAST and Rotated BRIEF): an efficient alternative to SIFT. Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP). Feature matching using BFMatcher. Introduction to camera models: pinhole camera model, intrinsic and extrinsic parameters. Optical flow: Lucas-Kanade method — concept and applications.

Unit 5	Deep Learning for Image Classification and Object Detection	9 hours

Overview of popular CNN architectures: **AlexNet, VGGNet, ResNet**. Transfer learning: concept of fine-tuning pre-trained models for image classification. Introduction to object detection: **sliding window approach, HOG+SVM detector**. Overview of region-based CNNs (**R-CNN, Fast R-CNN, Faster R-CNN**). Single-shot detectors: **YOLO (You Only Look Once)**-architecture and working principle. Data augmentation strategies and their role in improving generalization.

Total Lecture hours	45 hours

Textbook:

1. Rafael C. Gonzalez, Richard E. Woods, Digital Image Processing, Pearson Education, 4th Edition, 2018.
2. S. Jayaraman, S. Essakkirajan, T. Veerakumar, Digital Image Processing, Tata McGraw Hill (TMH).
3. Adrian Rosebrock, Deep Learning for Computer Vision with Python, PyImageSearch, 2017.

Reference Books

1. Milan Sonka, Vaclav Hlavac, Roger Boyle, Image Processing, Analysis, and Machine Vision, Cengage Learning.
2. Richard Szeliski, Computer Vision: Algorithms and Applications, Springer, 2nd Edition, 2022.
3. Simon J.D. Prince, Computer Vision: Models, Learning and Inference, Cambridge University Press, 2012.
4. Digital Image Processing and Computer Vision - Sonka, Hlavac, Boyle, Cengage Learning.

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
60	15	75	150
75			

Course Code: HS205L	Course Name: Aptitude - 4	L	T	P	C
		1	0	0	1

Pre-requisite: NIL

Course Objectives:

1. To gain a strong foundation in basic mathematical concepts, and become proficient in performing calculations, solving equations, and manipulating numerical data.
2. To apply quantitative methods effectively in interpreting data, optimizing solutions, and making informed decisions in academic, professional, and real-world engineering contexts.
3. To build a strong foundation in basic mathematical concepts including arithmetic, algebra, geometry, mensuration, trigonometry, and data interpretation required for engineering and analytical problem-solving.
4. To prepare for various competitive exams, such as aptitude tests for job placements, college admissions tests for higher education (GATE/CAT/GRE/GMAT) and government entrance exams. The course will provide the necessary knowledge and practice to perform well in these exams.

Course Outcome: After completion of the course, the student will be able to

1. gain a strong foundation in mathematical concepts and demonstrate proficiency in calculations, equation solving, and numerical data manipulation, with the ability to **apply** these skills to solve mathematical and engineering problems.
2. apply the principles of permutation, combination, and probability to **analyze** uncertainty and solve counting and chance-based problems relevant to mathematical and engineering applications.
3. Interpret and **analyse** data presented in various formats such as tables, graphs, and charts to support statistical analysis and informed decision-making.
4. **Analyze &** apply geometric principles and trigonometric relationships to analyze shapes, angles, and measurements; solve problems involving distances, heights, areas, and volumes; interpret diagrams and spatial data; and use these concepts effectively in mathematical, scientific, and engineering applications; quantitative and logical aptitude concepts through gamified learning techniques to solve problems with improved accuracy, speed, and confidence, while demonstrating enhanced analytical thinking, decision-making, and competitive readiness for academic and placement assessments.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	1		1		1					2
CO2	3	1		1		1					2
CO3	3	1		1		1					2
CO4	3	1		1		1					2

Unit1	Number Systems-II and Algebra	3 hours

Types of numbers: (Surds indices, Logarithms, Progressions, Factors, Factorizations, Number of Zeros, Algebraic Identities, Linear and Quadratic Equations.

Unit 2	Permutation, Combination and Probability	3 hours

Set theory, Definition and Introduction of permutation and combination, Fundamental principle of counting by "AND" or "OR" rule, Forming of Numbers, Words, and Team; Problems related to linear and nonlinear arrangement. Application based on Geometry, Match, handshake, Chess board, Whole number, and Natural number solution, etc. Distribution: - Identical & Distinct, Basic concepts, definition and terminology related to probability, Problem based on Dice, Coins, Playing cards, Marbles, etc. Conditional Probability – Bayes theorem.

Unit 3	Data Interpretation, Data Sufficiency and Statistics	3 hours

Data Interpretation: Tables (Simple & Complex), Bar Graphs, Line Graphs, Pie Charts, Histograms, Radar graph, Caselets, Mixed Charts, Missing Data Problems, Comparison & Ratio Analysis, Percentage Change Analysis, Time-Based Data Sets, Data Sufficiency.

Statistics: Measures of Central Tendency (Mean, Median, Mode), Measures of Dispersion (Range, Mean Deviation, Variance, Standard Deviation), Averages & Weighted Mean, Quartiles, Deciles & Percentiles.

Unit 4	Geometry, Mensuration and Trigonometry	3 hours

Geometry: Basics of Geometry: Lines and Angles, Triangles (Types, Properties, Theorems), Congruence of Triangles, Similarity of Triangles, Quadrilaterals (Parallelogram, Rectangle, Square, Rhombus, Trapezium), Polygons, Circles (Chord, Tangent, Secant, Angles in a Circle). **Mensuration** – 2D Figures (Area & Perimeter), Mensuration – 3D Figures (Surface Area & Volume). **Trigonometry:** Introduction to Trigonometry, Trigonometric Ratios (sin, cos, tan, cosec, sec, cot), Trigonometric Identities, Values of Trigonometric Functions (Standard Angles), Trigonometric Equation, Heights and Distances.

Unit 5	Flowcharts and Gamified Aptitude	3 hours

Number Puzzles, Crypt Arithmetic, Flowcharts, Geo-Sudoku, Digit Challenge, Switch Challenge, Motion Challenge, Grid Challenge.

	Total Lecture hours	15 hours

Textbook

1. Booklet provided by the Aptitude Team, Training Division -CRPC, KIET

Reference Books

1. "Quantitative Aptitude for Competitive Examinations" by R.S. Aggarwal, S. Chand Publication.
2. "How to Prepare for Quantitative Aptitude for the CAT" by Arun Sharma, TMH Publication.
3. "Magical Book on Quicker Maths" by M. Tyra, Published by BSC Publishing Co. Pvt. Ltd
4. "Quantitative Aptitude Quantum CAT" by Sarvesh K. Verma, Arihant Publication.

Reference Websites:

1. https://www.indianbix.com
2. https://www.prepinsta.com
3. https://www.talentsprint.com
4. https://www.careeride.com
5. https://www.smartkeeda.com

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	25	25	50
25			

Open Elective-I Courses Detail Syllabus – VI Sem

Course Code: MA2020	Course Name: Reliability Theory and Maintenance	L	T	P	C
		2	0	0	2

Pre-requisite: 10+2

Course Objectives:

1. The objective of this course is to familiarize the graduate engineers with the concept of Reliability.
2. To equip students with techniques for reliability modeling, maintenance planning, and optimization of system availability and maintainability.

Course Outcome:

1.Understand the concept of Reliability of system with failure and distribution function.
2. Apply the concept of reliability on redundant system.
3. Employ the concept of reliability on complex system.
4. Understand the knowledge of Modeling, Maintainability and Availability of the system

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	2				1				2
CO2	2	2	2				1				2
CO3	2	2	2				1				2
CO4	2	2	2				1				2

Unit 1	Introduction to Reliability	08 hours

Definition of Reliability, Concept of Failure (Failure, failure rate, hazard rate), Types of failures: Early failure (infant mortality), Random failure, Wear-out failure; Failure density function, Bathtub Curve, Mean time to failure, Mean Time To Repair (MTTR), Mean Time Between Failures (MTBF), Estimation of Reliability function from failure data. Probability Distributions in Reliability: Binomial distribution, Poisson distribution, Exponential distribution, Weibull distribution.

Unit 2	System Reliability	08 hours

Configuration of System components: Series Configuration, Parallel Configuration, Mixed configurations, k-out-of-n:G/F Configuration, Definition of Redundancy, Types of Redundancy: Active redundancy and Standby (passive) redundancy; Reliability evaluation of the system (Series systems, Parallel systems, Mixed system); Reliability of k-out-of-n:G/F system, reliability of Standby system (with perfect and imperfect switching).

Unit 3	Reliability of Complex Systems	07 hours

Definition of Complex system, Decomposition method, Conditional Probability approach, Minimal Path set method, Cut set method, Tie set method, Reliability Based Optimization, Introduction to Stochastic FEM

Unit 4	Modeling, Maintainability and Availability	07 hours

Reliability Block Diagrams (RBD), Fault tree analysis (FTA), Life Testing and Estimation: Parameter estimation (MLE method) and Goodness-of-fit tests, Markov models, Maintainability function, Availability function

Total hours	30 hours

Textbook:

1: E. Balaguruswamy, “Reliability Engineering”, Tata McGraw Hill.
2: D J Smith, Reliability, Maintainability & Risk, Butterworth-Heinemann

Reference Books:

1: Charles E Ebeling, “Reliability and Maintainability Engineering, Tata McGraw Hill
2: R.Billintan and R.N. Allan, "Reliability Evaluation of Engineering and Systems", Plenum Press.
3: K.C. Kapoor& L.R. Lamberson, "Reliability in Engineering and Design", John Wiely and Sons.
4: G.H.Sandler, "System Reliability Engineering", Prentice Hall.

Mode of Evaluation:

MSE	CA	ESE	Total
40	10	50	100

Continuous Assessment (CA): Video Assignments/ Assignment/ Quiz/ Project/ Seminar/ Group Discussion/ MOOC Course etc.

Course Code: EC2370	Course Name: Introduction to Robotic Systems	L	T	P	C
		2	0	0	2

Course Objectives:

1. To understand fundamentals of robotic systems, components and configurations.
2. To analyze robot kinematics, including forward/inverse kinematics and coordinate transformations.
3. To develop mathematical models of robot manipulators using DH parameters, Jacobian, and dynamic approaches.
4. To apply AI techniques (ANN, RL, Computer Vision) in robotic systems for perception, control, and decision-making.

Course Outcome: After completion of the course, the student will be able to

1. Explain and classify different types of robots, components, DOF and configurations.
2. Apply forward and inverse kinematics and perform coordinate transformations using homogeneous matrices.
3. Develop kinematic and dynamic models of robotic manipulators using DH parameters, Jacobian, and dynamic equations, and analyze system behavior.
4. Implement AI-based solutions such as: Object detection using OpenCV & YOLO and ANN-based control and trajectory generation.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	2	-	-	2	-	2	2	-	2
CO2	2	2	2	-	-	2	-	2	2	-	2
CO3	3	3	3	-	-	3	-	3	3	-	2
CO4	3	2	2	-	3	3	3	2	2	3	3

Unit 1	Introduction to Robotic Systems	7 hours

Fundamentals of Robotics: Types of Robots, Robot Components, Manipulator, Reference Frames, Degrees of Freedom, Joints of a Robot, Robot Configurations, End Effector.

Unit 2	Kinematic Transformations	7 hours

Kinematics Equations: Forward Kinematics & Inverse Kinematics, Workspace

Coordinate Frames and Homogeneous Transformation: Coordinate Transformations, Rotation, Translation, Composition of homogeneous transformations

Unit 3	Kinematic Model for Robot Manipulator	8 hours

Denavit-Hartenberg Procedure & Coordinates Frames, 3-DOF & 6-DOF Manipulator, 4-Axis SCARA Robot Manipulator Jacobian

Dynamics of Manipulator: Lagrange-Euler approach, Newton-Euler approach, Multiple Degree of Freedom

Unit 4	AI-Integrated Robotic Systems	8 hours

Introduction to AI in Robotics: Role of AI in modern robotics, Difference between Traditional vs Intelligent robots

AI for Robot Perception: Sensor data processing, Basics of computer vision using OpenCV, Object detection

Total Lecture hours	30 hours

Textbook

1. Subir Kumar Saha, “Introduction to Robotics”, Tata McGraw-Hill Education, 2014.
2. Richard Paul, Robot Manipulators: Mathematics, Programming and Control, MIT Press, 1981
3. Robert Shilling, Fundamentals of Robotics, Prentice-Hall , 2003
4. Laxmidhar Behera and Indrani Kar, “Intelligent Systems and Control”, Oxford University Press, Nov 2009.

Reference Books

1. M. Felix Orlando, Ashish Dutta, Anupam Saxena, Laxmidhar Behera, Tomoya Tamei and Tomohiro Shibata, "Manipulability Analysis of Human Thumb, Index and Middle Finger in Cooperative 3D Rotational Movements of a Small Object", Robotica, vol. 31, pp. 797-809, 2013.
2. M. Felix Orlando, Laxmidhar Behera, Tomayo Tamei, Tomohiro Shibata, Ashish Dutta and Anupam Saxena," On Redundancy Resolution of the Human Thumb, Index and Middle Fingers in Cooperative Object Translation," Robotica, vol. 35, pp. 1992-2017, 2016.

Mode of Evaluation:

MSE	CA	ESE	Total
40	10	50	100

CA1- Report Submission on Addverb Technologies Visit**, CA2-**Assignment Based**, CA3-**Attendance

Course Code: EE214O	Course Name: AIoT for Smart Energy	L	T	P	C
		2	0	0	2

Pre-requisite: Basic knowledge of Electronics, Programming in Python/C, and Mathematics.

Course Objectives:

1. To introduce the fundamentals of Artificial Intelligence of Things (AIoT), its architecture in the context of smart energy systems and to design IoT sensor networks for real-time energy monitoring, data acquisition, edge-level processing.
2. To develop the ability to apply machine learning and AI algorithms for energy demand forecasting, anomaly detection, renewable energy prediction and smart grid technologies, cloud platforms, and the future roadmap of AIoT in sustainable energy.

Course Outcome: After completion of the course, the student will be able to

1. Understand the fundamentals of AIoT architecture, communication protocols, and their role in the smart energy ecosystem.
2. Design and analyze IoT sensor networks for real-time energy monitoring and data acquisition.
3. Apply machine learning and AI algorithms to predict energy demand and detect anomalies in energy systems.
4. Evaluate smart grid technologies, cloud platforms, renewable energy integration, and AIoT-based optimization strategies.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	2	2	2	-	-	-	-	-	2
CO2	3	2	2	2	3	-	-	-	-	-	3
CO3	3	2	2	2	3	-	-	-	-	-	3
CO4	3	2	2	2	3	-	-	-	-	-	3

Unit 1	Foundations of AIoT and Smart Energy	7 hours

Overview of AIoT: convergence of Artificial Intelligence and Internet of Things. AIoT system architecture: Perception, Network, Platform and Application layers. Introduction to smart energy systems, smart meters and Advanced Metering Infrastructure (AMI). Key energy metrics: kWh, Power Factor and Load Factor. Communication protocols: MQTT, Zigbee and LoRaWAN. Edge computing versus cloud computing in energy applications.

Unit 2	IoT Sensors, Devices and Connectivity	8 hours

Energy sensors: current transformers, voltage sensors, irradiance sensors and smart plugs. Microcontroller platforms: Arduino and ESP32 for energy monitoring. Wireless communication: Zigbee and Bluetooth for short range, LoRaWAN for long range, NB-IoT for smart grid applications. Introduction to Edge AI: TinyML and TensorFlow Lite for on-device processing. Basic data pre-processing: noise filtering, handling missing values and time-series data characteristics.

Unit 3	Artificial Intelligence for Energy Management	8 hours

Overview of supervised and unsupervised machine learning with energy examples. Energy demand forecasting using Linear Regression, Random Forest and LSTM networks. Anomaly detection for electricity theft and equipment faults using Isolation Forest and Autoencoders. Solar and wind power prediction using weather data and ML models. Introduction to Reinforcement Learning for smart thermostat and battery scheduling optimization.

Unit 4	Smart Grids, Cloud Platforms and Future Applications	7 hours

Smart grid architecture: evolution from traditional grid to smart grid, demand response and Virtual Power Plants. Cloud platforms: AWS IoT Core, Google Cloud IoT and open-source tools such as Node-RED and ThingSpeak. AIoT for building energy management and EV smart charging. Introduction to Vehicle-to-Grid (V2G) technology. Emerging trends: Federated Learning for energy data privacy, Blockchain for peer-to-peer energy trading and role of AIoT in achieving Net Zero targets.

	Total Lecture Hours	30 hours

Textbook:

1. A. Bahga and V. Madisetti, Internet of Things: A Hands-On Approach, VPT, 2014.
2. F. Chollet, Deep Learning with Python, Manning Publications, 2021.
3. J. Ekanayake, K. Liyanage, J. Wu, A. Yokoyama, N. Jenkins, Smart Grid: Technology and Applications, Wiley, 2012.
4. A. Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 3rd ed., O'Reilly Media, 2022.

Reference Books:

1.S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 4th ed., Pearson, 2020.
2.R. H. Shumway and D. S. Stoffer, Time Series Analysis and Its Applications, Springer, 2017.
3.M. Schwartz, Internet of Things with ESP8266, Packt Publishing, 2016.
4.P. J. Sadalage and M. Fowler, NoSQL Distilled, Addison-Wesley, 2012.

Mode of Evaluation:

MSE	CA	ESE	Total
40	10	50	100

Course Code: ME2290	Course Name: Advanced Engineering Materials	L	T	P	C
		2	0	0	2

Pre-requisite: Basic knowledge of Physics and Chemistry

Course Objectives:

1. To introduce students to the classification and properties of advanced engineering materials.
2. To develop understanding of modern materials used in aerospace, biomedical, electronics, and energy applications.
3. To familiarize students with Semiconductors,smart materials, nanomaterials, and composite materials.
4. To enable students to correlate material properties with engineering applications and sustainability requirements.

Course Outcomes:

1. Explain the structure, properties, and classification of advanced engineering materials.
2. Analyze the applications of composite, smart, nano, and biomaterials in modern engineering systems.
3. Select suitable advanced materials for specific industrial and technological applications.
4. Understand sustainable and emerging material technologies for future engineering challenges.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	-	-	-	-	-	-	2
CO2	2	3	2	1	1	-	-	-	-	-	2
CO3	2	3	3	1	2	-	-	-	-	-	2
CO4	2	2	1	-	-	1	1	-	-	-	3

Unit 1	Composite materials	7 hours
Introduction to composite materials and their Classification. Fiber reinforced composites (FRP, CFRP, GFRP) Metal Matrix Composites (MMC), Ceramic Matrix Composites (CMC), Manufacturing methods and applications Properties and applications of composites in automotive and aerospace sectors.		
Unit 2	**Semiconductor and Smart Materials**	**8 hours**
Introduction to conductors, semiconductors, and insulators, Crystal structure of semiconductors, Energy band theory, Intrinsic and extrinsic semiconductors, Carrier concentration, mobility, conductivity, Fermi level concept Smart materials: shape memory alloys, piezoelectric materials, magnetostrictive materials, and electroactive materials. Applications of smart materials in sensors and actuators.		
Unit 3	**Nanomaterials and Biomaterials**	**8 hours**
Introduction to nanomaterials and nanotechnology. Synthesis and characterization of nanomaterials. Carbon nanotubes, graphene, nanoparticles, and Nano coatings. Applications of nanomaterials in electronics, energy storage, and environmental engineering. Biomaterials: metallic, ceramic, polymeric, and composite biomaterials. Applications in medical implants and tissue engineering.		
Unit 4	**Advanced Functional Materials and Sustainable Materials**	**7 hours**
Semiconductors and electronic materials. Superconducting materials and optical materials. Materials for renewable energy systems such as solar cells, batteries, and fuel cells. Eco-friendly and sustainable materials. Recycling and green processing of materials. Recent trends and future scope of advanced engineering materials.		
Total Lecture Hours		**30 hours**

Textbook:

1. Callister, W.D. and Rethwisch, D.G., Materials Science and Engineering: An Introduction, Wiley India.
2. Raghavan, V., Materials Science and Engineering, Prentice Hall India.
3. Donald R. Askeland and Wendelin J. Wright, The Science and Engineering of Materials, Cengage Learning.
4. William F. Smith and Javad Hashemi, Foundations of Materials Science and Engineering, McGraw Hill.

Reference Books:

1. Charles P. Poole Jr. and Frank J. Owens, Introduction to Nanotechnology, Wiley.
2. M.F. Ashby, H. Shercliff, and D. Cebon, Materials: Engineering, Science, Processing and Design.
3. S.O. Kasap, Principles of Electronic Materials and Devices, McGraw Hill.
4. K.K. Chawla, Composite Materials: Science and Engineering, Springer.

Mode of Evaluation:

MSE	CA	ESE	Total
40	10	50	100

Course Code: MS2030	Theory Course Name: Business Strategy	L	T	P	C
		2	0	0	2

Pre-requisite: NIL

Course Objectives

1. To develop strategic thinking from a business perspective
2. To analyse industry dynamics and strategic positioning.
3. To enable students to design business strategies in technology-driven environments.

Course Outcome (COs)

CO No.	Statement of Course Outcome
After the completion of the course, the student will be able to:	
CO 1	**Apply** strategic tools to business scenarios
CO 2	**Analyze** competitive environments and firm positioning
CO 3	**Evaluate** strategic choices and business models
CO 4	**Assess** the impact of innovation and emerging technologies on business strategy.

CO-PO Mapping (scale 1: low, 2: Medium, 3: High)

Course Outcomes (COs)	Programme Outcomes (POs)					Additional Programme Outcomes (APOs)	
	PO1	PO2	PO3	PO4	PO5	APO 1	APO 2
CO 1	1	2	2	2	2	3	3
CO 2	3	3	3	3	3	3	3
CO 3	2	3	3	3	2	3	3
CO 4	3	3	3	3	3	3	3

DETAILED SYLLABUS

Unit1	Foundations of Business Strategy	Hours 7

Strategy vs Business Model, Levels of strategy, Strategic management process, Concept of Competitive Advantage, Stakeholders in Business Strategy.

Unit2	Industry and Competitive Analysis	Hours 8

Industry Structure and Dynamics, Porter's Five Forces Model, Competitor Analysis , Market Segmentation and Targeting (basic concepts), Strategic Positioning, Tools: SWOT Analysis, PESTLE

Unit3	Strategic Choices and Business Models	Hours 8

Generic Strategies: Cost Leadership vs Differentiation, Focus Strategy (Niche markets), Growth Strategies: Expansion, Diversification, Innovation, Role of Technology and AI in shaping business strategy.

Unit4	Strategy Implementation & Contemporary Issues	Hours 7

Strategy Implementation: Organizational Structure and Culture, Leadership and Change Management, Resource Allocation. Contemporary Strategic Issues: Digital Transformation, Artificial Intelligence in Strategy, Sustainability and Corporate Social Responsibility (CSR), Ethics in Strategic Decision-Making.

Total Lecture hours	30 ours

Textbook

1. David & David – Strategic Management: Concepts and Cases
2. Michael Porter – Competitive Strategy
3. Osterwalder & Pigneur – Business Model Generation
4. Selected case studies (Harvard/industry-based simplified versions)

Reference Books

1. https://www.ebookbou.edu.bd/Books/Text/SOB/MBA/mba_3311/Unit-07.pdf
2. https://nptel.ac.in/courses/110/108/110108047/
3. https://swayam.gov.in/nd2_imb19_mg08/preview

Mode of Evaluation

MSE	CA	ESE	Total
40	**10**	50	100

Continuous Assessment (CA): Case Presentation/ Activities/ Assignment / Quiz / Project/ MOOC etc.

Professional Elective-III Courses Detail Syllabus – VI Sem

Course Code: CS320E	Course Name: Microservices Architecture & System Design	L	T	P	C
		2	0	4	4

Pre-requisite: Backend Servers & Api Development with Databases

Course Objectives:

1. To understand microservices architecture and distributed systems.
2. To design scalable and fault-tolerant systems.
3. To implement communication, API Gateway, and containerization.
4. To deploy and monitor microservices applications.

Course Outcome: After completion of the course, the student will be able to

1. Understand microservices architecture and principles
2. Design system architecture using HLD and LLD
3. Implement API Gateway and service communication
4. Apply messaging, caching, and rate limiting techniques
5. Deploy, scale, and monitor microservices systems

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	-	2	-	-	-	-	-	2
CO2	3	3	2	2	3	-	-	2	1	-	2
CO3	2	3	2	2	3	-	-	2	1	-	2
CO4	2	3	3	2	3	-	-	2	1	-	2
CO5	3	3	3	3	3	-	-	3	2	1	3

Unit 1 | Microservice Design and Fundamentals | 18 hours

Monolithic vs. Microservice Architectures, Principles of Microservices (Loose Coupling, Scalability, Fault Isolation), Domain-Driven Design (DDD) and Bounded Contexts, Synchronous vs. Asynchronous Communication, Designing RESTful APIs for Microservices, Database per Service: Data Partitioning and Consistency, Configuration and Service Discovery (Consul / etcd), Observability Basics: Logs, Metrics, and Health Checks, Node.js, Express, Consul, Postman, Git

Sample Problems:

1.Design and document a monolithic application architecture and then decompose it into microservices using a domain-driven approach.

2.Create a simple RESTful microservice using Node.js and Express that exposes CRUD APIs for a single domain (e.g., users or products).

3.Implement two microservices (e.g., users and orders) that communicate synchronously via REST APIs.

Unit 2 | System Design – HLD and LLD | 18 hours

Understanding System Design Fundamentals, High-Level Design (HLD): Architecture Diagrams, Components, and Data Flow, Low-Level Design (LLD): Classes, APIs, and Database Schemas, Design Patterns for Microservices: Repository, Event-Driven, Circuit Breaker, CQRS, Designing for Scalability, Availability, and Fault Tolerance, Trade-offs in System Design (CAP Theorem), Load Estimation, Latency, and Throughput Planning, Case Study: Designing a Scalable E-Commerce Platform, Draw.io, Lucidchart, Visual Studio Code

Sample Problems:

1. Design: a) HLD for online shopping system b) LLD with database schema
2. Build an asynchronous event-driven communication channel between microservices using RabbitMQ queues.

Unit 3 | API Gateway, Communication, and Containerization | 18 hours

Role of API Gateways in Microservices, API Gateway Patterns: Routing, Aggregation, Caching, Authentication, Building a Lightweight API Gateway using NGINX, RESTful Communication using Express.js and Axios, Containerization with Docker Community Edition, Writing Dockerfiles and Building Containers, Managing Multi-Container Apps using Docker Compose, Securing APIs with JWT and HTTPS, NGINX, Docker CE, Express.js, Axios

Sample Problems:

1. Design a service discovery mechanism using Consul or etcd and register multiple microservices for lookup.

2. Set up an API Gateway using NGINX to route incoming client requests to the appropriate microservices.
3. Secure microservice endpoints by implementing JWT-based authentication and authorization middleware.

Unit 4	Messaging, Caching, and Rate Limiting	18 hours

Asynchronous Messaging in Microservices, Message Broker Concepts and Event-Driven Architecture, Implementing Queues and Topics using RabbitMQ, Pub/Sub Model Implementation for Service Communication, Caching and Session Management using Redis, Redis Data Structures and Key Expiration, Implementing Rate Limiting and Request Throttling using Express Middleware, Token Bucket and Leaky Bucket Algorithms in Node.js, RabbitMQ OSS, Redis OSS, Express-rate-limit

Sample problems:

1. Containerize multiple microservices using Docker and orchestrate them using Docker Compose.
2. Implement rate limiting and request throttling using Express middleware to protect APIs from abuse.

Unit 5	Deployment, Scaling, and Load Balancing	18 hours

Deployment Strategies: Rolling, Blue-Green, Canary, Container Orchestration with Docker Compose, Scaling Microservices using Multiple Containers, Load Balancing Techniques: Round Robin and Least Connections, Implementing Load Balancing with NGINX Reverse Proxy, Continuous Integration & Delivery using GitHub Actions, Monitoring with Prometheus and Grafana, Logging and Tracing with ELK Stack (Elasticsearch, Logstash, Kibana), Capstone: Deploy a Load-Balanced Microservice Application on Docker, Docker CE, NGINX, GitHub Actions, Prometheus, Grafana, ELK Stack

Sample problems:

1. Configure load balancing for microservices using NGINX reverse proxy and test traffic distribution under load.
2. Monitor running microservices using Prometheus and visualize metrics (CPU, memory, requests) in Grafana dashboards.
3. Implement centralized logging for all microservices using the ELK Stack (Elasticsearch, Logstash, Kibana).
4. Design, build, and deploy a fully functional microservices-based application (e.g., food delivery or e-commerce) with API Gateway, Redis cache, RabbitMQ, Docker Compose deployment, Prometheus monitoring, and NGINX load balancing.

Total Lecture hours	90 hours

Textbook

1. Building Microservices: Designing Fine-Grained Systems – Sam Newman
2. Microservices Patterns – Chris Richardson
3. Designing Data-Intensive Applications – Martin Kleppmann

Reference Books

1. Microservices: Up and Running – Ronnie Mitra & Irakli Nadareishvili
2. Production-Ready Microservices – Susan J. Fowler

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: CS309E	Course Name: Agentic AI: Building Intelligent & Self-Learning Systems	L	T	P	C
		2	0	4	4

Pre-requisite: Machine Learning Fundamentals, Python Programming, Natural Language Processing Basics

Course Objectives:

1. To introduce students to the concepts, scope, and evolution of Agentic AI and its role in modern intelligent systems.
2. To enable students to distinguish between traditional AI and agent-based systems and understand their architectures.
3. To develop understanding of key agentic AI components: autonomy, reasoning, memory, and tool use.
4. To explore advanced agentic AI frameworks such as AutoGPT, Devin AI, and OpenAI Assistants API.
5. To cultivate awareness of ethical considerations, responsible AI principles, and governance in autonomous AI development.

Course Outcome: After completion of the course, the student will be able to

1. Understand the definition, scope, evolution, and characteristics of Agentic AI and analyze its applications in real-world intelligent systems.
2. Compare and contrast traditional AI architectures with agent-based systems, analyzing their strengths, limitations, and use cases.
3. Design and implement agentic AI components including autonomy mechanisms, reasoning strategies, memory modules, and tool integrations.
4. Build and evaluate agentic AI applications using frameworks such as AutoGPT, Devin AI, and OpenAI Assistants API for real-world tasks.
5. Apply ethical frameworks and responsible AI principles to identify and mitigate bias, ensure fairness, accountability, and transparency in autonomous AI systems.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	2	-	-	-	-	-	2
CO2	3	3	2	-	2	-	-	2	-	-	2
CO3	3	3	3	2	3	-	1	2	-	-	2
CO4	3	3	3	2	3	-	1	2	-	-	2
CO5	2	2	2	-	1	3	2	2	1	-	2

Unit 1	Definition and Scope of Agentic AI	18 hours

Introduction to Agentic AI, Definition and characteristics of Agentic AI, Difference between AI and Agentic AI, Evolution of Agentic AI, Scope of Agentic AI in modern applications, Understanding autonomy in Agentic AI, Key goals of Agentic AI systems, Role of agents in decision-making processes, Types of agentic behaviors, Agentic AI in real-world applications, Comparison of Agentic AI with traditional AI systems, Benefits of Agentic AI in business and tech, Limitations and challenges of Agentic AI, Role of Agentic AI in autonomous systems, Future trends in Agentic AI.

Sample Problems:

1. Research and present a comparative analysis of traditional AI and Agentic AI in a real-world application domain (e.g., healthcare, autonomous vehicles, or customer service). Identify five agentic behaviors exhibited and justify how autonomy is achieved.
2. Design a conceptual framework for an Agentic AI system targeting a modern business challenge. Define the agents' goals, scope, decision-making roles, and describe its potential benefits and limitations in that context.

Unit 2	Traditional AI vs Agent-Based Systems	18 hours

Defining Traditional AI, The concept of agent-based systems, Key differences between traditional AI and agent-based systems, Reactive vs. deliberative agents, Architecture of traditional AI, Architecture of agent-based systems, Use cases for traditional AI, Use cases for agent-based systems, How traditional AI solves problems, How agent-based systems solve problems, Pros and cons of traditional AI, Pros and cons of agent-based systems.

Sample Problems:

1. Implement a basic reactive agent and a deliberative agent in Python to solve a navigation problem (e.g., a grid-based environment). Compare their problem-solving approaches, response times, and adaptability to changing environments.
2. Design and build a simple agent-based simulation (e.g., a traffic management or supply-chain model) contrasting its architecture with an equivalent traditional AI rule-based system. Analyze the trade-offs in scalability and flexibility.

Unit 3	Key Components: Autonomy, Reasoning, Memory, Tool	18 hours

Understanding autonomy in AI, Types of autonomy in Agentic AI, The importance of reasoning in agent-based systems, Types of reasoning: deductive, inductive, and abduction, Memory in Agentic AI, Types of memory in AI systems, How memory impacts decision-making, Tools used by agentic AI, Role of tools in enhancing agent capabilities, How agents use tools to achieve goals, The link between reasoning and autonomy, Self-adaptation in autonomous systems.

Sample Problems:

1. Develop a memory module for an agentic AI system in Python that supports both short-term and long-term memory. Demonstrate how different memory types influence decision-making in a simulated task environment (e.g., a conversational assistant that recalls user preferences across sessions).
2. Implement a reasoning engine that applies deductive, inductive, and abductive reasoning to solve a real-world problem (e.g., medical diagnosis or fault detection). Integrate a tool-use mechanism so the agent can call external APIs or knowledge bases to enhance its reasoning capability.

Unit 4	AutoGPT, Devin AI, OpenAI Assistants API	18 hours

Introduction to AutoGPT, Capabilities of AutoGPT, How AutoGPT generates autonomous text, Use cases for AutoGPT, Introduction to Devin AI, Features of Devin AI, How Devin AI utilizes agentic behavior, Use cases of Devin AI in industries, Overview of OpenAI Assistants API, Functionalities of OpenAI Assistants API, Comparing AutoGPT with OpenAI API, Application of OpenAI Assistants in customer service, Real-world use cases of AutoGPT and Devin AI, Challenges faced by AutoGPT and Devin AI, Future directions of AutoGPT and Devin AI.

Sample Problems:

1. Set up and run AutoGPT to autonomously complete a multi-step task (e.g., researching a topic, writing a report, and saving it). Document its chain-of-thought, tool usage, and decision points. Compare this workflow to the same task completed using the OpenAI Assistants API.
2. Build a customer service chatbot using the OpenAI Assistants API with custom functions/tools (e.g., checking order status, generating FAQs). Evaluate its performance against an AutoGPT-based equivalent on task completion rate, accuracy, and user experience.

Unit 5	Ethical Considerations & Responsible AI Development	18 hours

Introduction to AI ethics, Ethical challenges in Agentic AI, Privacy concerns in autonomous systems, Fairness in Agentic AI, Transparency in agent decision-making, Ensuring accountability in autonomous agents, Bias in agent-based systems, Addressing discrimination in AI systems, The role of human oversight in Agentic AI, Safety measures in the development of Agentic AI, Responsibility for AI's actions, Legal implications of autonomous decision-making, Impact of Agentic AI on employment, Environmental considerations in Agentic AI, Future of ethical frameworks for Agentic AI.

Sample Problems:

1. Investigate a real-world case study involving an ethical failure in an AI or autonomous system (e.g., biased hiring algorithm, autonomous vehicle incident). Analyze the root causes, stakeholder impact, and propose a redesigned system with transparency, fairness, and accountability mechanisms built in.
2. Design an ethically responsible agentic AI system for a sensitive domain (e.g., healthcare decision support or criminal justice). Define the human oversight mechanisms, bias detection methods, privacy safeguards, and legal compliance requirements, and implement a prototype with fairness checks.

Total Lecture Hours	90 hours

Textbook:

1. Artificial Intelligence: A Modern Approach (4th Edition), Stuart Russell and Peter Norvig, Pearson
2. Building LLM Apps with LangChain, Ben Auffarth, Packt Publishing
3. Hands-On Large Language Models, Jay Alammar and Maarten Grootendorst, O'Reilly Media

Reference Books:

1. Multi-Agent Systems: Algorithmic, Game-Theoretic, and Logical Foundations, Yoav Shoham and Kevin Leyton-Brown, Cambridge University Press
2. The Age of AI: And Our Human Future, Henry Kissinger, Eric Schmidt and Daniel Huttenlocher, Little, Brown and Company
3. Designing Autonomous AI: Principles and Practices for Building Ethical Intelligent Systems, O'Reilly Media

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: CS306E	Course Name: Cloud & Infrastructure Management	L	T	P	C
		2	0	4	4

Pre-requisite: AI Powered CI/CD & Containerization (Linux, GitHub, Docker, Jenkins, Kubernetes, CI/CD)

Course Objectives:

1. To develop a strong understanding of cloud architecture, multi-cloud strategies, and infrastructure automation using modern tools like Terraform, Ansible, and cloud platforms.
2. To enable learners to design, deploy, monitor, and optimize intelligent cloud systems using AI-driven operations (AIOps), FinOps practices, and secure LLM-based applications.

Course Outcome: After completion of the course, the student will be able to

1. Design and implement cloud solutions using IaaS, PaaS, SaaS models across platforms like Amazon Web Services, Microsoft Azure, and Google Cloud Platform.

2. Apply Infrastructure as Code (IaC) tools such as Terraform and Ansible to automate provisioning, configuration, and compliance of cloud resources.
3. Analyze system performance using observability tools and implement AI-driven monitoring, incident detection, and automated remediation strategies.
4. Optimize cloud costs and resource utilization using FinOps principles, predictive analytics, and intelligent scaling techniques.
5. Design secure, scalable cloud architectures by integrating AI/LLM applications and implementing automated security, compliance, and governance mechanisms.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	1	2	-	-	-	1	-	1
CO2	2	2	3	1	3	-	-	1	1	-	1
CO3	2	3	2	2	3	-	-	1	1	-	1
CO4	2	2	2	2	3	1	-	-	1	2	1
CO5	2	2	3	1	3	2	2	1	1	1	2

Unit 1 | Cloud Architecture & Multi-Cloud Strategy | 18 hours

Cloud computing models (IaaS, PaaS, SaaS), AWS / , Azure / GCP comparative overview Networking & identity in cloud environments, Multi-cloud & hybrid cloud approaches (Anthos, Azure Arc), Provision multi-cloud VMs and deploy apps with Terraform

SampleProblems:

1. Provision cloud infrastructure (VMs, networks) using Terraform or Pulumi on AWS/Azure/GCP.

Unit 2 | Infrastructure Automation & Configuration Management | 18 hours

Infrastructure as Code in depth (Terraform, Ansible, Pulumi), Automated provisioning and drift correction, AI-assisted configuration recommendations (Ansible Autopilot, Chef Automate), Continuous compliance monitoring using AI policies, Implement IaC workflow with AI-driven config validation

Sample Problems:

1. Automate server configuration using Ansible to install, configure, and validate services remotely.
2. Implement AI-assisted configuration management using Ansible Autopilot or Chef Automate AI.

Unit 3 | Intelligent Monitoring & AIOps | 18 hours

Monitoring vs Observability: logs, metrics, traces, Tools: Datadog AI, Dynatrace Davis AI, Moogsoft, AI-driven incident correlation and root cause analysis, Automated alerting and auto-remediation (PagerDuty AI, Opsgenie AI), Configure AI-based observability dashboard and alert workflow

Sample Problems:

1. Deploy Datadog AI or Dynatrace to collect real-time performance metrics and detect anomalies automatically.
2. Configure the ELK Stack to centralize logs and use AI-based correlation to identify incidents faster.
3. Enable distributed tracing with OpenTelemetry for a microservices-based cloud application.
4. Integrate Moogsoft or Opsgenie AI to automate alert correlation and incident prioritization.
5. Configure PagerDuty AI for automated incident response and self-healing workflows.

Unit 4 | Cloud FinOps & Predictive Optimization | 18 hours

Cloud cost optimization strategies and budgeting, Predictive scaling using historical metrics + ML models, Automated cost alerts using AWS Cost Anomaly Detection, Kubecost, Intelligent resource allocation with serverless AI (AWS Lambda, GCP AI Functions), Build a FinOps dashboard to forecast and optimize cloud spend

Sample problems:

1. Build a predictive auto-scaling model using ML to forecast server load and optimize resource utilization.
2. Implement a FinOps dashboard using AWS Cost Anomaly Detection or Kubecost for cloud cost forecasting.
3. Develop a Lambda or Cloud Function that triggers automatic resource recovery after a system failure.

Unit 5	Security, Compliance & LLM Ops Integration	18 hours

Cloud security automation (Wiz.io, Prisma Cloud AI), AI-powered risk assessment and policy as code, Integrating LLM applications into cloud pipelines (LLM Ops), Deploying and monitoring LLMs with vLLM or Hugging Face Endpoints, Deploy a cloud-hosted LLM with auto-scaling and AI-based compliance monitoring

Sample problems:

1. Use Wiz.io or Prisma Cloud AI to continuously scan and fix cloud security misconfigurations.
2. Deploy an LLM-based chatbot (Llama 2 / Falcon) on the cloud and integrate observability and scaling.
3. Create an AIOps-enabled autonomous cloud system that monitors, self-heals, and optimizes cost and security in real time.

Total Lecture Hours	90 hours

Textbook:

1. Cloud Computing: Concepts, Technology & Architecture – Thomas Erl
2. Infrastructure as Code – Kief Morris

Reference Books:

1. Site Reliability Engineering – Google
2. AI Engineering – Chip Huyen

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: CS338E	Course Name: Advanced Data Engineering & AWS Cloud Solutions	L	T	P	C
		2	0	4	4

Pre-requisite: AWS Cloud Architecture and Data Engineering (5th Semester)

Course Objectives:

The objective of this course is to advance learners from professional-grade cloud architecture competencies to expert-level, enterprise-scale data engineering and machine learning engineering practices using Amazon Web Services (AWS). Students will engineer petabyte-scale data lakehouse and big-data processing platforms, build and operationalize MLOps pipelines for production machine learning and generative AI workloads, implement federated data governance and data mesh architectures, automate data pipelines through DataOps and CI/CD practices, and design secure, compliant, multi-cloud data strategies — while preparing for the AWS Certified Data Engineer – Associate (DEA-C01) certification

Course Outcome: After completion of the course, the student will be able to

1. Design and implement large-scale distributed data processing and data lakehouse architectures using Apache Spark, Amazon EMR, and open table formats.
2. Build, deploy, and operationalize end-to-end machine learning and generative AI pipelines using Amazon SageMaker, Amazon Bedrock, and MLOps best practices.
3. Architect enterprise data governance frameworks, including data mesh, federated analytics, and advanced metadata and data-quality management.
4. Apply DataOps principles to automate, test, and monitor data pipelines through CI/CD, infrastructure as code, and advanced observability practices.
5. Design secure, compliant, and cost-optimized multi-cloud and hybrid data platforms that meet enterprise regulatory and business requirements.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	1	2	-	-	-	-	-	2
CO2	3	3	3	2	3	1	-	-	-	-	2
CO3	3	3	2	2	3	1	1	-	-	-	2
CO4	3	2	3	3	3	2	-	-	-	-	2
CO5	3	3	2	2	3	2	2	-	-	-	2

Unit 1	Big Data Processing, Distributed Computing, and Data Lakehouse Engineering	18 hours

Distributed Computing Foundations and Apache Spark on AWS

- Spark architecture: driver, executors, DAG scheduler, and Catalyst optimizer
- RDDs, DataFrames, and Datasets; lazy evaluation and partitioning strategies
- Spark performance tuning: shuffle optimization, broadcast joins, and caching strategies

Amazon EMR – Managed Big Data Platform

- EMR cluster architecture: primary, core, and task nodes; instance fleets and groups
- EMR Serverless: application configuration, auto-scaling, and cost-efficient job execution
- EMR on EKS: running Spark workloads on Kubernetes; integration with existing container platforms

Data Lakehouse Architecture

- Apache Iceberg, Apache Hudi, and Delta Lake on AWS: ACID transactions, time travel, and schema evolution
- AWS Glue integration with lakehouse table formats; querying via Athena and Redshift Spectrum
- Lakehouse vs. traditional data lake vs. data warehouse: architectural trade-offs

Batch Processing Orchestration at Scale

- AWS Batch: compute environments, job queues, and array jobs for large-scale ETL
- AWS Glue workflows and job orchestration for complex multi-step pipelines
- Performance benchmarking and cost modelling for large-scale batch workloads

Case study: Building a Petabyte-Scale Data Lakehouse with Apache Iceberg and EMR for a Retail Analytics Platform

Unit 2	Machine Learning Engineering, MLOps, and AI-Driven Data Pipelines	18 hours

Amazon SageMaker – End-to-End ML Platform

- SageMaker Studio, notebooks, and built-in algorithms; bring-your-own-container models
- SageMaker Pipelines: model training, evaluation, and deployment workflows
- SageMaker Feature Store: online and offline stores, feature groups, and point-in-time queries

Model Deployment and Inference Architecture

- Real-time endpoints, serverless inference, and asynchronous inference patterns
- Multi-model endpoints and inference pipelines for cost-efficient model serving
- Model monitoring: data drift, model quality, and bias detection with SageMaker Model Monitor

MLOps and Automation

- CI/CD for ML: SageMaker Projects, model registry, and approval workflows
- SageMaker Pipelines integration with Step Functions and EventBridge
- Experiment tracking, model versioning, and reproducibility practices

Generative AI and Foundation Models on AWS

- Amazon Bedrock: foundation model access, fine-tuning, and knowledge bases
- Retrieval-Augmented Generation (RAG) architectures using OpenSearch and Bedrock
- Responsible AI: guardrails, bias mitigation, and governance for generative AI systems

Case Study: Designing an MLOps Pipeline for Real-Time Fraud Detection Using SageMaker and Feature Store

Unit 3	Enterprise Data Governance, Data Mesh, and Federated Analytics	18 hours

Data Mesh Architecture on AWS

- Domain-oriented data ownership, data products, and self-serve data infrastructure
- Federated governance models: balancing domain autonomy with organization-wide standards
- Implementing data mesh with Lake Formation, Glue Data Catalog, and cross-account sharing

Advanced Data Cataloging and Metadata Management

- AWS Glue Data Catalog at scale: catalog federation and hybrid metadata stores
- Amazon Data Zone: data portal, business glossary, and data product publishing
- Data lineage tracking and impact analysis across complex pipelines

Data Quality and Observability Engineering

- AWS Glue Data Quality: rule-based validation, anomaly detection, and quality scorecards
- Data observability patterns: freshness, volume, schema, and distribution monitoring
- Building automated data quality gates within CI/CD pipelines

Federated and Cross-Account Analytics

- Cross-account data sharing with Lake Formation and Redshift data sharing
- Query federation across S3, Redshift, RDS, and on-premises sources

Multi-tenant analytics architectures with row-level and column-level security

Case Study: Implementing an Organization-Wide Data Mesh with Federated Governance for a Multi-Business-Unit Enterprise

Unit 4	**DataOps, Pipeline Automation, and Advanced Observability**	**18 hours**

DataOps Principles and Practices

- DataOps lifecycle: continuous integration, testing, deployment, and monitoring of data pipelines
- Version control strategies for data pipelines, schemas, and infrastructure
- Testing data pipelines: unit tests, integration tests, and data contract validation

CI/CD for Data Engineering Workloads

- Infrastructure as Code for data platforms: CDK and CloudFormation patterns for Glue, EMR, and Redshift
- Automated deployment pipelines for Glue jobs, Lambda functions, and Step Functions state machines
- Blue-green and canary deployment strategies for data pipeline updates

Advanced Orchestration Patterns

- Amazon MWAA at scale: DAG optimization, dynamic task mapping, and cross-DAG dependencies
- Event-driven pipeline architectures using EventBridge, SQS, and SNS for decoupled processing
- Error handling, retries, dead-letter queues, and idempotency in distributed pipelines

Observability and Performance Engineering

- Advanced CloudWatch: custom metrics, embedded metric format, and anomaly detection
- Distributed tracing across data pipelines with AWS X-Ray and OpenTelemetry
- Cost observability: tagging strategies, AWS Cost Explorer, and chargeback/showback models

Case Study: Building a Fully Automated DataOps Platform with CI/CD, Testing, and Observability for a Financial Services Data Team

Unit 5	**Multi-Cloud Data Strategy, Advanced Security, and Certification Preparation**	**18 hours**

Hybrid and Multi-Cloud Data Architecture

- Interoperability patterns: AWS with Azure and GCP data services; avoiding vendor lock-in
- Data replication and synchronization strategies across cloud providers
- Multi-cloud orchestration using open-source tools (Apache Airflow, Terraform) alongside native AWS services

Advanced Security for Data Platforms

- Fine-grained access control: Lake Formation tag-based access control at scale
- Data encryption architectures: envelope encryption, client-side encryption, and key rotation strategies
- Securing ML pipelines: model artifact encryption, endpoint access control, and data privacy in training

Enterprise Compliance and Risk Management

- Advanced compliance frameworks: SOC 2, FedRAMP, and industry-specific regulations
- Automated compliance monitoring with AWS Config rules and Security Hub custom insights
- Data residency, sovereignty, and cross-border data transfer considerations

AWS Certified Data Engineer – Associate (DEA-C01) Exam Domain Review

- Domain 1 – Data Ingestion and Transformation (34%): streaming, batch, and transformation services
- Domain 2 – Data Store Management (26%): storage, cataloging, and lifecycle management
- Domain 3 – Data Operations and Support (22%): orchestration, monitoring, and troubleshooting
- Domain 4 – Data Security and Governance (18%): access control, encryption, and compliance

Case Study: Architecting a Secure, Multi-Cloud, Compliant Data Platform for a Global Financial Institution

Total Lecture Hours	**90 hours**

Textbook:

1. Damji, J., Wenig, B., Das, T., & Lee, D. (2023). Learning Spark: Lightning-Fast Data Analytics (2nd ed.). O'Reilly Media.
2. Sarkar, T. (2023). AWS Certified Data Engineer Study Guide: Associate DEA-C01 Exam. Sybex / Wiley.

Reference Books:

1. Reis, J., & Housley, M. (2022). Fundamentals of Data Engineering. O'Reilly Media.
2. Huyen, C. (2022). Designing Machine Learning Systems. O'Reilly Media.
3. AWS Documentation and Whitepapers: AWS Well-Architected Framework Data Analytics Lens, AWS Machine Learning Lens, and AWS Data Mesh Guidance.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
80	**20**	**100**	**200**
100			

Course Code: CS323E	**Course Name: Advanced iOS Frameworks & App Store Deployment**	**L**	**T**	**P**	**C**
		2	**0**	**4**	**4**

Pre-requisite: iOS App Development & Swift UI

Course Objectives:

1.To develop Advanced User Interface Design and Native Hardware Frameworks.
2. To build and Implement Robust Local Data Persistence and System Notifications.

Course Outcomes:

1.To apply interactive SwiftUI interfaces using animations, gestures, and custom view modifiers.
2. To Develop iOS applications using Core Data and Realm for local data storage and management.
3. To apply audio and video playback features into SwiftUI applications.
4. To develop location-based services and interactive maps using Core Location and MapKit.
5. To Develop and debug a capstone iOS application by integrating UI enhancements, local databases, maps, media, and notifications.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	3		3	-	-	-	-	-	2
CO2	3	3	3	2	3	-	-	-	-	-	2
CO3	2	2	3		3	-	-	-	-	-	2
CO4	3	3	3	2	3	-	-	-	-	-	2
CO5	3	3	3	2	3	-	-	-	-	-	2

Unit 1	SwiftUI UI/UX Enhancements	18 hours

Basic Animations: Implementing simple implicit and explicit view animations.
Transitions & Gestures: Adding basic view transitions and handling common interactions like Tap, Drag, and Long Press.
Custom View Modifiers: Creating simple, reusable modifiers to apply consistent styles across views.

Unit 2	Local Databases - Core Data & Realm	18 hours

Core Data Essentials: Setting up simple entities and performing basic CRUD (Create, Read, Update, Delete) operations.
Introduction to Realm: Setting up a Realm database as a lightweight alternative to Core Data.
Data Saving: Saving and fetching local user data within SwiftUI views.

Unit 3	Basic Media & Audio-Video Playback	18 hours

Media Playback: Using basic framework tools to play audio and video files inside an app.
Playback UI: Building straightforward play/pause controls in SwiftUI.
Media Management: Importing and displaying basic audio/video assets cleanly.

Unit 4	Location Services & Maps	18 hours

Core Location Basics: Requesting location permissions and getting the user's current GPS coordinates.
MapKit Integration: Displaying a basic interactive map inside a SwiftUI view.
Map Annotations: Adding simple pins or markers to show specific locations on the map.

Unit 5	Notifications & Capstone Project	18 hours

Local Notifications: Triggering simple scheduled alerts and reminders on the user's device.
Project Planning: Combining UI enhancements, local databases, maps, and notifications into a single app idea.
Final Implementation: Building and debugging a simplified capstone application on the simulator.

Total Lecture Hours	90 hours

Textbook:

1. iOS App Development Essentials (SwiftUI Edition) by Neil Smyth
2. SwiftUI Cookbook by Giordano Scalzo and Edgar N. Miranda

Reference Books:

1. Pro iPhone Development with SwiftUI: Designing Professional Apps with Apple's Frameworks by Wallace Wang (Apress)
2. AVFoundation Framework Reference by Apple Documentation & Developer Publications

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: IT308E	Course Name: AWS Security Specialty	L	T	P	C
		2	0	4	4

Pre-requisite: Cloud Computing Fundamentals, Basic Networking, Basic Cyber Security Concepts

Course Objectives: This course provides comprehensive knowledge of AWS security services and best practices aligned with the AWS Certified Security Specialty certification. It focuses on securing AWS workloads through identity and access management, data protection, infrastructure security, incident response, and monitoring. The course aims to develop practical skills for designing secure cloud architectures, implementing compliance controls, detecting threats, and preparing students for AWS security certification and industry roles.

Course Outcome:

CO1: Understand AWS security principles, shared responsibility model, and secure cloud architecture design.
CO2: Implement identity and access management using IAM, roles, policies, and multi-factor authentication.
CO3: Configure infrastructure security including network security, compute protection, and logging mechanisms.
CO4: Apply data protection strategies including encryption, key management, and secure storage.
CO5: Monitor, detect, and respond to security incidents using AWS security services and prepare for AWS Security Specialty certification.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	2	-	1	-	-	-	2
CO2	3	3	2	2	3	-	1	-	-	-	2
CO3	3	3	2	2	3	-	1	-	-	-	2
CO4	3	2	2	2	3	-	1	-	-	-	2
CO5	3	3	2	2	3	-	1	-	-	-	2

Unit 1	AWS Identity and Access Management and Security Fundamentals	18 hours

AWS shared responsibility model, security best practices, IAM users, groups, roles, policies, and permissions. Multi-Factor Authentication (MFA), identity federation, AWS Organizations, service control policies (SCPs), and secure account management.

Case Study: Securing AWS Accounts using IAM Roles, Policies, and MFA.

Unit 2	Infrastructure Security and Network Protection	18 hours

Amazon VPC security, subnets, route tables, security groups, network ACLs. AWS WAF, AWS Shield, AWS Firewall Manager. Securing EC2 instances, patch management, and secure configuration practices.

Case Study: Designing a Secure VPC Architecture with WAF and Shield.

Unit 3	Data Protection and Encryption	18 hours

Data encryption at rest and in transit, AWS Key Management Service (KMS), Cloud HSM, secrets management using AWS Secrets Manager. Securing S3, EBS, RDS, and database encryption mechanisms.

Case Study: Implementing End-to-End Data Encryption using AWS KMS.

Unit 4	Security Monitoring and Incident Response	18 hours

AWS CloudTrail, CloudWatch, AWS Config. Threat detection using Amazon Guard Duty, AWS Security Hub, Amazon Inspector. Incident response strategies, logging, alerting, and automated remediation.
Case Study: Threat Detection and Automated Incident Response using Guard Duty and Security Hub.

Unit 5	Governance, Risk, Compliance, and Advanced Security	18 hours

AWS compliance programs (ISO, GDPR basics), AWS Artifact, auditing and governance. Secure DevOps practices, infrastructure as code security, vulnerability management, and best practices for secure cloud architecture.
Case Study: Implementing Compliance and Governance using AWS Config and AWS Artifact.

Total Lecture Hours	90 hours

Textbook:

1. Zeal Vora – AWS Certified Security Specialty Study Guide
2. Stuart Scott – AWS Certified Security Specialty Exam Guide
3. AWS Documentation – Security Best Practices

Reference Books:

1. AWS Training and Certification Portal
2. https://aws.amazon.com/training/

AWS Security Specialty Certification Guide

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Practical Courses Detail Syllabus – VI SEM

Course Code: HS206P	Course Name: Soft Skills for Employability - 2	L	T	P	C
		0	0	2	1

Pre-requisite: Successful completion of Soft Skills Essentials I & II and Employability Skills – I (Verbal Ability).

Course Objectives:

To strengthen students' placement preparedness and develop employability skills by enhancing creativity, critical thinking, resume development, resume preparation, group discussion skills, and interview techniques.

Course Outcome: After completion of the lab, the student will be able to

CO1: Demonstrate interpersonal effectiveness through handling group discussions and professional presence through visumes.

CO2: Justify their resumes through effective professional presentation skills.

CO3: Apply interview techniques and feedback to successfully handle employment interviews.

CO4: Develop a professional digital identity by utilizing AI-enabled career tools, networking platforms, psychometric insights, and employability strategies to enhance career readiness.

CO-PO Mapping (scale 1: low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	1	-	2	3	-	2
CO2	-	-	-	-	-	-	-	-	3	-	2
CO3	-	-	-	-	-	-	-	1	3	-	2
CO4	-	-	-	-	-	1	-	-	2	-	2

List of Activities:

1. LLM Enabled Professional Self-Introductions using Mind-Mapping
2. Employability & Placement Process Overview
3. Resume Review Lab – Peer Review & Faculty Feedback
4. Group Discussion Practice Sessions
5. Personal Branding for Career Development – LinkedIn, AlmaConnect, E-Portfolio
6. Psychometric Test (MBTI / Locus of Control)
7. Preparation for Job Interviews – 4 Ts (Types, Tips, Tricks & Techniques)
8. Pair & Peer Interview Practice
9. Practising HR, Behavioural and Situational Questions
10. Handling HR Questions (Extempore)
11. Company Specific & Profile Specific Interviews (AI Tools)
12. Online Spoken English Test

Total Laboratory Hours: 30 hours

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	50	-	50

Course Code: CS317P	Course Name: Problem Solving Lab-2	L	T	P	C
		0	0	2	1

Pre-requisite: Knowledge of Data Structure

Course Objectives:

1. To familiarize students with the basic and advance of Problem Solving.
2. To learn modern techniques to solve DSA Problems.

Course Outcome:

Course Outcomes: After completion of the lab, the student will be able to

1.Apply data-driven and combinatorial problem-solving techniques

2.Design solutions using Dynamic Programming

3.Develop solutions using trees and graph algorithms

4. Integrate advanced techniques to solve complex problems

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	1	2	-	-	-	-	-	1
CO2	3	3	3	1	2	-	-	-	-	-	1
CO3	3	3	3	2	2	-	-	-	-	-	1
CO4	2	3	3	2	3	-	-	2	1	1	2

List of Experiments:

1. Hashing Basics- 1. Frequency counting 2. First non-repeating character 3. Two sum using hashing
2. Hash Map + Sets Advanced -1. Group anagrams 2. Longest consecutive sequence 3. Subarray sum = k
3. Prefix Sum / Difference Array - 1. Range sum queries 2. Rainwater trapping 3.Kadane's algorithm
4. Recursion +Backtracking Advanced- 1. Sudoku solver 2. Word search in grid 3. Combination sum
5. Dynamic Programming– I (Introduction) - 1 . Fibonacci DP 2. Climbing stairs 3. Min cost climbing stairs
6. DP – II (1D DP) 1. House robber 2. Decode ways 3. Partition equal subset sum
7. DP – III (2D DP) 1. Longest common subsequence 2. Edit distance 3. 0/1 knapsack
8. DP – IV (Advanced) 1. Matrix chain multiplication 2. Palindromic subsequence 3. LIS
9. Trees Basics(Conceptual Only)- 1. Preorder traversal 2. Height of tree 3. BFS on tree
10. Binary Search Tree Techniques - 1. Validate BST 2. LCA 3. Kth smallest element
11. Graphs – BFS/DFS 1. Number of islands 2. Bipartite graph check 3. Connected components
12. Graphs – Shortest Path 1. Dijkstra 2. Bellman–Ford 3. Shortest path in grid
13. Graphs – Advanced 1 . Topological sort 2. Course schedule 3. Detect cycle in directed graph
14. Mixed Techniques / DSA Integration - 1. Sliding window + hashing problem 2. Graph + BFS+ DP hybrid 3. String + DP complex case
15. Final Mini Project 1. 20-question problem sheet 2. Contest style evaluation 3. Full pattern-based lab exam

Total Laboratory Hours: 30 hours

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	50	-	50
50			